Estimation of Large Latent Factor Models for Time Series Data *

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We use the factor model and a procedure in the same spirit to Pan and Yao (2008) for utilizing autocorrelation information in multivariate time series data to carry out dimension reductio. The dimension of the data, p, can be larger than the sample size n. Estimators for the unknown factors and the corresponding factor loadings matrix are obtained. We show that when all the factors are strong in the sense that the norm of each column in the factor loading matrix is of the order $p^{1/2}$, the estimated factor loadings matrix, as well as the estimated precision matrix for the original data, converge weakly in L_2-norm to the true ones at a rate independent of p. This result demonstrates clearly when the "curse" is canceled out by the "blessings" in dimensionality. Theoretical results are also developed when not all factors are strong. We also give theoretical comparisons of two procedures in doing so, indicating advantages of a two-step procedure.

We also show that the method cannot estimate the covariance matrix better than the sample covariance matrix, which coincides with the result in Fan et al. (2008) when the factors are known. Our theoretical results are further illustrated by a simulation study and an application to a real data set.

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1 Introduction

Analysis of large data sets is becoming an integral part of study across many scientific disciplines. High dimensional multiple time series is among the many challenging data sets that are important to fields like finance, economics, environmental studies, and many others. For example, asset returns of a relatively (to the sample size) large number of assets are important to asset allocation. Environmental time series are often of high dimension because of the large number of variables monitored. A common parsimonious model like the VARMA(p, q) is not viable without regularization since the number of parameters grows with the square of the dimension of the time series. Hence it is important to reduce the dimension of the data before we make further analysis.

Factor model is a traditional tool for this purpose, see for example Anderson (1963), Priestley and Tong (1974), Brillinger (1981) and Peña and Box (1987) for some earlier theoretical works. Since analysis of high dimensional multiple time series is becoming more important, there are more recent papers which consider the dimension p of the time series growing with the sample size n. Bai (2003) presented asymptotic distributions of various quantities for a factor model with unknown factors; Forni et al. (2002) focused on the dynamic factor model in forecasting, and Fan et al. (2008) analyzed a factor model with known factors, with rates of convergence specified on the resulting covariance and inverse covariance matrices, all allowing p to grow with n.

In this paper we focus on a new methodology which, in the same spirit to Peña and Poncela (2006) or Pan and Yao (2008), explores the autocorrelation of the time series data for the estimation of the factor loadings matrix and the factors in a factor model. By considering the autocorrelation of the data, we can outperform least squares-based method like that in Bai and Ng (2002), as demonstrated empirically in our simulations section. However our method is also different from Peña and Poncela (2006) or Pan and Yao (2008) in many ways. It is based on principle component analysis of a matrix constructed from the autocovariance matrices of the data. Yet, unlike Peña and Poncela (2006), our method does not require the calculation of the inverse of the sample covariance matrix of the data, which is invertible only when $p < n$, and is in general a bad estimator of the population covariance matrix when p is comparable or even larger than n. Indeed we analyze the rates of convergence of various estimators of the factor model allowing p to grow with n, which is an important generalization to Peña and Poncela (2006) and Pan and Yao (2008) where they assumed p is fixed. Moreover, we still allow for correlations between 'past' noise and 'future' factors, and make no distributional assumptions in the model. We consider stationary factors in this paper, which is an important step towards generalization to non-stationary factors, and is done in Lam et al. (2010) due to the amount of materials involved.

We also generalize the factor model to allow for strong or weak factors, see for example Chudik et al. (2009). However, unlike their definition, we explicitly characterize the strength of a factor by a parameter δ, which is more convenient to present how the strength of a factor affects any rates of convergence. It is an important generalization since the presence of factors with different strength can affect the rate of convergence of our estimation, as shown in section 3 and thereafter. A two-step procedure is introduced to remedy this, and is shown both theoretically and empirically that it can outperform the original procedure, especially in the presence of weak factors.

The paper is organized as follows. Section 2 presents the factor model with assumptions, and describes the estimation methodology. The identification of a particular version of the factor loadings matrix is explained in the appendix. Theories on the rates of convergence for the factor loadings matrix, the covariance matrix and the precision matrix are presented in section 3 when the factors are of the same strength. These results are then extended to combinations of factors with different strength. Extensive simulations are done in section 5, with analysis of a set of implied volatility data given in section 6. All proofs are relegated to section 7.

2 Models and Estimation Methodology

2.1 The factor model

Let $\mathbf{y}_1, \cdots, \mathbf{y}_n$ be n $p \times 1$ successive observations from a vector time series process. The factor model assumes

$$\mathbf{y}_t = \mathbf{A}\mathbf{x}_t + \boldsymbol{\epsilon}_t, \tag{2.1}$$

where $\mathbf{x}_t$ is a $r \times 1$ time series of unobserved factors with finite second moments, which is assumed stationary in this paper (non-stationary factors are considered in Lam et al. (2010)). $\mathbf{A}$ is a $p \times r$ unknown constant factor loadings matrix, and $r \leq p$ is the number of factors. The white noise process $\{\boldsymbol{\epsilon}_t\}$ has mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$. Our main assumptions in this paper are as follows:

(I) $\text{Cov}(\boldsymbol{\epsilon}_t, \mathbf{x}_s) = \mathbf{0}$ for all $s \leq t$.

(II) $\text{Cov}(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_s) = \mathbf{0}$ for all $s \neq t$.

(III) There exists no linear combination of the elements of $\mathbf{x}_t$ which is a white noise process.

Assumption (I) means that future noise and past factors are uncorrelated, whereas assumption (II) means that the noise series is serially uncorrelated. These two assumptions allow us to develop our methodology to utilize information from the autocorrelations of the observed time series in estimating the factor model. Assumption (III) is needed to maintain the validity of assumption (I) when $s = t$. If it does not hold, such a linear combination of $\mathbf{x}_t$ should become part of $\boldsymbol{\epsilon}_t$ and the assumption will hold again. Assumptions for the factor loadings matrix $\mathbf{A}$ and the factors $\mathbf{x}_t$ will be presented in section 2.2.

The above model has been considered by various authors, for instance Peña and Box (1987). The dynamic factor model, proposed by Chamberlain and Rothschild (1983) and further studied by, among others, Forni et al. (2002), can be written in this form too with each component of $\mathbf{x}_t$ being an moving average process. See also Bai and Ng (2007) for further details.

In this paper, we assume that the number of factors r is known. It remains an open challenge to extend our theoretical results to the case with r unknown. There is a large body of literature on determining r in such a factor model. For instance Bai and Ng (2002) developed a general information criterion for consistently estimating r, and proposed some different penalty functions in practice. Bai and Ng (2007) also developed tests for determining the number of shocks in a dynamic factor model, while Hallin and Liška (2007) presented an information criterion for determining the number of shocks for a more general form of a dynamic factor model. Bathia et al. (2009) utilized a bootstrap approach for determining r. Pan and Yao (2008) adopted a simple sequential test procedure in determining r. In section 5, we use an information criterion from Bai and Ng (2002) to determine r in simulations requiring comparisons of different estimation procedures.

2.2 Identifiability

Model (2.1) is not identifiable, since for any invertible $r \times r$ matrix $\mathbf{H}$, we can replace $\mathbf{A}$ by $\mathbf{AH}$ and $\mathbf{x}_t$ by $\mathbf{H}^{-1}\mathbf{x}_t$ without altering it. Note that this does not change the factor loading space, i.e. the linear space spanned by the columns of $\mathbf{A}$. In order to identify $\mathbf{A}$, a common practice is to set $\mathbf{A}^T\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\mathbf{A}$ to be a diagonal matrix. We treat the issue differently and we will identify explicitly the version of the factor loadings matrix to be estimated, which is a natural choice resulted from the methodology presented in section 2.3.

Let $\boldsymbol{\Sigma}_{\mathbf{x}}(k) = \mathrm{Cov}(\mathbf{x}_{t+k}, \mathbf{x}_t)$, $\quad \boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k) = \mathrm{Cov}(\mathbf{x}_{t+k},\ \boldsymbol{\epsilon}_t)$, and

$$\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k) = (n-k)^{-1}\sum_{t=1}^{n-k}(\mathbf{x}_{t+k} - \bar{\mathbf{x}})(\mathbf{x}_t - \bar{\mathbf{x}})^T,$$
$$\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}}(k) = (n-k)^{-1}\sum_{t=1}^{n-k}(\mathbf{x}_{t+k} - \bar{\mathbf{x}})(\boldsymbol{\epsilon}_t - \bar{\boldsymbol{\epsilon}})^T,$$

with $\bar{\mathbf{x}} = n^{-1}\sum_{t=1}^n \mathbf{x}_t$, $\bar{\boldsymbol{\epsilon}} = n^{-1}\sum_{t=1}^n \boldsymbol{\epsilon}_t$. $\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}(k)$ and $\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon},\mathbf{x}}(k)$ are defined similarly. Denote by $\|M\|$ the spectral norm of M, which is the positive square root of the maximum eigenvalue of MM^T; and by $\|M\|_{\min}$ the positive square root of the minimum eigenvalue of MM^T or M^TM, whichever has a smaller matrix size. The notation $a \asymp b$ represents $a = O(b)$ and $b = O(a)$. Some regularity conditions are now in order.

(A) The factor process $\{\mathbf{x}_t\}$ is stationary (to be extended to non-stationary factors in Lam et al. (2010)). For $k = 0, 1, \cdots, k_0$, where $k_0 \geq 1$ is a small positive integer, we have
$$\|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\| \asymp 1 \asymp \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|_{\min}.$$
The cross autocovariance matrix $\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)$ has elements of order $O(1)$.

(B) $\mathbf{A} = (\mathbf{a}_1 \cdots \mathbf{a}_r)$ such that $\|\mathbf{a}_i\|^2 \asymp p^{1-\delta}$, $\ i = 1, \cdots, r$, $\ 0 \leq \delta \leq 1$.

(C) For each $i = 1, \cdots, r$ and δ given in (B), $\ \min_{\theta_j, j\neq i}\|\mathbf{a}_i - \sum_{j\neq i}\theta_j\mathbf{a}_j\|^2 \asymp p^{1-\delta}$.

Assumption (A) is essentially saying that the elements in the factor series $\mathbf{x}_t$ have non-trivial autocorrelations at lag k, in the sense that the maximum autocovariance between the elements are exactly non-zero constant, and no linear combination of the elements of $\mathbf{x}_t$ are white noise or asymptotically white noise, which corresponds to assumption (I) in section 2.1. Since we control the strength of factors, which can grow with n, via the factor loadings matrix $\mathbf{A}$, assuming constant maximum autocovariance in $\mathbf{x}_t$ does not lose any generality.

When $\delta = 0$ in assumption (B), the corresponding factors are called strong factors since it includes the case where each element in $\mathbf{a}_i$ is $O(1)$, implying that the factors are shared (strongly) among the majority of the p cross-sectional variables. On the other hand, they are called weak factors when $\delta > 0$. Weak and strong factors are not new concepts. Chudik et al. (2009) introduced a notion of strong and weak factors through the finiteness of the average expected absolute values of the factor loadings along a column of the factor loadings matrix. In this paper, we use a parameter δ to represent the strength of a factor which gives more explicit rates of convergence and shows clearer how dimensionality plays a role when the factors can be weak. Assumption (C) implies that no single $\mathbf{a}_i$ can be represented as a linear combination of other $\mathbf{a}_j, j \neq i$ (see also (B)). Otherwise there should be fewer than r factors with $\|\mathbf{a}_i\|^2 \asymp p^{1-\delta}$, and there should be other weaker factors (larger δ) in the model.

To facilitate our estimation, it is important to normalize the factor loadings matrix, so that $\mathbf{A}$ becomes orthogonal with $\mathbf{A}^T\mathbf{A} = \mathbf{I}_r$. This can be achieved by the QR decomposition. See section 7 for the derivation. The model then becomes

$$\begin{aligned} \mathbf{y}_t = \mathbf{A}\mathbf{x}_t + \boldsymbol{\epsilon}_t, \quad \mathbf{A}^T\mathbf{A} = \mathbf{I}_r, \quad \text{with} \\ \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\| \asymp p^{1-\delta} \asymp \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|_{\min}, \quad \|\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| = O(p^{1-\delta/2}), \\ \mathrm{Cov}(\mathbf{x}_s, \boldsymbol{\epsilon}_t) = \mathbf{0} \quad \text{for all } s \leq t, \\ \mathrm{Cov}(\boldsymbol{\epsilon}_s, \boldsymbol{\epsilon}_t) = \mathbf{0} \quad \text{for all } s \neq t. \end{aligned} \tag{2.2}$$

Unless specified otherwise, all $\mathbf{y}_t$, $\mathbf{x}_t$, $\boldsymbol{\epsilon}_t$ and $\mathbf{A}$ in the sequel are defined in (2.2).

2.3 Estimation

Our estimation procedure is based on the same idea as Pan and Yao (2008), although our implementation is much more efficient and can handle the case with large p. Observe that for $k \geq 1$, model (2.2) implies that

$$\boldsymbol{\Sigma}_{\mathbf{y}}(k) = \mathrm{Cov}(\mathbf{y}_{t+k}, \mathbf{y}_t) = \mathbf{A}\boldsymbol{\Sigma}_{\mathbf{x}}(k)\mathbf{A}^T + \mathbf{A}\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k). \tag{2.3}$$

Hence for any orthogonal complement $\mathbf{B}$ of $\mathbf{A}$ (i.e. $\mathbf{B}^T\mathbf{B} = \mathbf{I}_{p-r}$ and $\mathbf{B}^T\mathbf{A} = \mathbf{0}$), we have $\mathbf{B}^T\boldsymbol{\Sigma}_{\mathbf{y}}(k) = \mathbf{0}$ for $k \geq 1$. Now define $\mathbf{L}$ to be the matrix

$$\begin{aligned} \mathbf{L} &= \sum_{k=1}^{k_0} \boldsymbol{\Sigma}_{\mathbf{y}}(k)\boldsymbol{\Sigma}_{\mathbf{y}}(k)^T \\ &= \mathbf{A}\Big(\sum_{k=1}^{k_0} (\boldsymbol{\Sigma}_{\mathbf{x}}(k)\mathbf{A}^T + \boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k))(\boldsymbol{\Sigma}_{\mathbf{x}}(k)\mathbf{A}^T + \boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k))^T\Big)\mathbf{A}^T. \end{aligned} \tag{2.4}$$

Clearly $\mathbf{LB} = \mathbf{0}$. Hence the columns of $\mathbf{B}$ are all the eigenvectors of $\mathbf{L}$ corresponding to zero-eigenvalue. We can write $\mathbf{L} = \mathbf{A}\mathbf{Q}_x\mathbf{D}_x\mathbf{Q}_x^T\mathbf{A}^T$, with all eigenvalues of the diagonal matrix $\mathbf{D}_x$ non-zero ranging from order $p^{2-2\delta}$ to $p^{2-\delta}$. Hence $\mathbf{A}\mathbf{Q}_x^{\pm}$ (which is defined by $\mathbf{W}^{\pm} = (\pm\mathbf{w}_1 \ \cdots \ \pm\mathbf{w}_m)$ for any matrix $\mathbf{W} = (\mathbf{w}_1 \ \cdots \ \mathbf{w}_m)$) contains all the eigenvectors of $\mathbf{L}$ corresponding to all the non-zero eigenvalues, which can be taken as columns of our factor loadings matrix estimator $\widehat{\mathbf{A}}$.

Note that the definition of $\mathbf{L}$ is invariant under rotation of the factor loadings matrix and factors, i.e. replacing $\mathbf{A}$ by $\mathbf{AQ}$ and $\mathbf{x}_t$ by $\mathbf{Q}^T\mathbf{x}_t$ for an orthogonal matrix $\mathbf{Q}$, and $\mathbf{L}$ does not change. Hence WLOG, we assume that model (2.2) already has factor loadings matrix $\mathbf{A}$ and factors $\mathbf{x}_t$ rotated with $\mathbf{Q}_x$ such that $\mathbf{LA} = \mathbf{AD}_x$. This way we are directly estimating $\mathbf{A}$ by $\widehat{\mathbf{A}}$.

In practice k_0 is chosen to be a small integer to avoid accumulating too much errors in our estimator $\widetilde{\mathbf{L}}$ of $\mathbf{L}$, defined by

$$\widetilde{\mathbf{L}} = \sum_{k=1}^{k_0} \widetilde{\mathbf{\Sigma}}_{\mathbf{y}}(k)\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}(k)^T, \quad \widetilde{\mathbf{\Sigma}}_{\mathbf{y}}(k) = (n-k)^{-1}\sum_{t=1}^{n-k}(\mathbf{y}_{t+k} - \bar{\mathbf{y}})(\mathbf{y}_t - \bar{\mathbf{y}})^T, \tag{2.5}$$

with $\bar{\mathbf{y}} = n^{-1}\sum_{t=1}^{n}\mathbf{y}_t$. For the first r largest eigenvalues, the corresponding r orthonormal eigenvectors of $\widetilde{\mathbf{L}}$, say $\widehat{\mathbf{a}}_1, \cdots, \widehat{\mathbf{a}}_r$, form our estimator for the factor loadings matrix $\widehat{\mathbf{A}} = (\widehat{\mathbf{a}}_1 \cdots \widehat{\mathbf{a}}_r)$. Consequently, we estimate the factors and the residuals by respectively

$$\widehat{\mathbf{x}}_t = \widehat{\mathbf{A}}^T\mathbf{y}_t, \qquad \mathbf{e}_t = \mathbf{y}_t - \widehat{\mathbf{A}}\widehat{\mathbf{x}}_t = (\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)\mathbf{y}_t. \tag{2.6}$$

3 Theoretical Results

In this section we present the rates of convergence for the estimator $\widehat{\mathbf{A}}$ for model (2.2), as well as the corresponding estimators for the covariance matrix and the precision matrix, which utilize the factor model structure and the estimator $\widehat{\mathbf{A}}$. We need the following assumption:

(D) It holds for any $0 \le k \le k_0$ that the elementwise rates of convergence for $\widetilde{\mathbf{\Sigma}}_{\mathbf{x}}(k) - \mathbf{\Sigma}_{\mathbf{x}}(k)$, $\widetilde{\mathbf{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)$ and $\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}(k)$ are respectively $O_P(n^{-l_x})$, $O_P(n^{-l_{x\epsilon}})$ and $O_P(n^{-l_\epsilon})$, for some constants $0 < l_x, l_{x\epsilon}, l_\epsilon \le 1/2$. We also have, elementwise, $\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon},\mathbf{x}}(k) = O_P(n^{-l_{x\epsilon}})$.

This assumption requires only the elementwise convergence of the sample cross and autocovariances for the factors $\{\mathbf{x}_t\}$ and noise $\{\boldsymbol{\epsilon}_t\}$. But with this we can specify the rates of convergence for our estimators already.

Theorem 1 *Under model (2.2) and assumption (D), if* $\|\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| = o(\|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|)$*, it holds that*

$$\|\widehat{\mathbf{A}} - \mathbf{A}\| = O_P(h_n) = O_P(n^{-l_x} + p^{\delta/2} n^{-l_{x\epsilon}} + p^{\delta} n^{-l_\epsilon}),$$

where we assume $h_n = o(1)$*. In particular, if all the factors are strong (i.e.* $\delta = 0$*), the rate of convergence does not depend on* p*.*

This theorem shows explicitly how the strength of the factors (through the factor loadings) affects the rate of convergence. It converges faster as the factors become stronger (i.e. δ gets smaller). When $\delta = 0$, the rate is independent of p. This shows that the curse of dimensionality is exactly offset by the information from the cross-sectional data when the factors are strong. Note that this result does not need explicit constraints on the structure of $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$ and $\boldsymbol{\Sigma}_{\mathbf{x}}$ other than implicit constraints from assumption (D).

To present the rates of convergence for the covariance matrix estimator $\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}$ of $\boldsymbol{\Sigma}_{\mathbf{y}}$ and its inverse, we introduce more assumptions.

(M1) The error-variance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$ is diagonal, and can be written as (after appropriate ordering of the cross-sectional data points)

$$\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}} = \mathrm{diag}(\sigma_1^2 \mathbf{1}_{m_1}^T, \sigma_2^2 \mathbf{1}_{m_2}^T, \cdots, \sigma_k^2 \mathbf{1}_{m_k}^T),$$

for some $k \geq 1$, where all the σ_j^2 are uniformly bounded away from 0 and infinity as $n \to \infty$. Here $\mathbf{1}_{m_j}$ is a column vector of ones with length m_j.

(M2) Define $s = \min_{1 \leq i \leq k} m_i$ and h_n as in Theorem 1. Then $p^{1-\delta} s^{-1} h_n^2 \to 0$ and $r^2 s^{-1} < p^{1-\delta} h_n^2$.

Assumption (M1) requires that on top of a diagonal structure on the error-variance $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$, there are k groups of cross-sectional data points, each group has size m_j with equal error-variance σ_j^2, $j = 1, \cdots, k$. This facilitates a pooled estimator for $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$ which is consistent. Assumption (M2) is made solely for simplification of the resulting rates of convergence in the subsequent theorems. They can be relaxed at the expense of more complicated convergence rates.

We estimate $\boldsymbol{\Sigma}_{\mathbf{y}}$ by

$$\begin{aligned}&\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}} = \widehat{\mathbf{A}} \widehat{\boldsymbol{\Sigma}}_{\mathbf{x}} \widehat{\mathbf{A}}^T + \widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}, \quad \text{with} \quad \widehat{\boldsymbol{\Sigma}}_{\mathbf{x}} = \widehat{\mathbf{A}}^T (\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}} - \widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}) \widehat{\mathbf{A}} \quad \text{and} \\ &\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} = \mathrm{diag}(\widehat{\sigma}_1^2 \mathbf{1}_{m_1}^T, \widehat{\sigma}_2^2 \mathbf{1}_{m_2}^T, \cdots, \widehat{\sigma}_k^2 \mathbf{1}_{m_k}^T), \quad \widehat{\sigma}_j^2 = n^{-1} s_j^{-1} \|\Delta_j \widehat{\mathbf{E}}\|_F,\end{aligned} \tag{3.7}$$

where $\Delta_j = \mathrm{diag}(\mathbf{0}_{m_1}^T, \cdots, \mathbf{0}_{m_{j-1}}^T, \mathbf{1}_{m_j}^T, \mathbf{0}_{m_{j+1}}^T, \cdots, \mathbf{0}_{m_k}^T)$, $\widehat{\mathbf{E}} = (\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)(\mathbf{y}_1 \cdots \mathbf{y}_n)$, and the norm $\|M\|_F$ denotes the Frobenius norm, defined by $\|M\|_F = \mathrm{tr}(M^T M)^{1/2}$.

In practice we do not know the value of k and the grouping which divides the cross-sectional variables into the k groups. We may start with one single group, and estimate $\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} = \widehat{\sigma}^2 \mathbf{I}_p$. By looking at the resulting residuals series and the corresponding sample covariance matrix, we may group the variables with similar magnitude of variances together. We then fit the model again with the constrained covariance structure specified in (M1).

Theorem 2 *Under assumption (D), it holds that*

$$\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| = O_P(p^{1-\delta} h_n) = O_P(p^{1-\delta} n^{-l_x} + p^{1-\delta/2} n^{-l_{x\epsilon}} + p n^{-l_\epsilon}).$$

Moreover, if in addition the assumptions from Theorem 1, and (M1) and (M2) also hold, then $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}$ *defined in (3.7) for model (2.2) has*

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| = O_P(p^{1-\delta} h_n).$$

This theorem tells us that asymptotically there is no difference in using the sample covariance matrix or the factor model-based covariance matrix estimator in (3.7) even when the factors are strong. In such a case both estimators have rates of convergence linear in p. This result is in line with Fan et al. (2008) where they have shown the sample covariance matrix as well as the factor model-based covariance matrix estimator are consistent in Frobenius norm at a rate linear in p, with the factors known in advance. This is numerically illustrated in section 5.

When concerning the precision matrix $\mathbf{\Sigma}_{\mathbf{y}}^{-1}$, the performance of the estimator $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1}$ is significantly better than the sample counterpart.

Theorem 3 *Under the assumptions in Theorem 1, (M1) and (M2), it holds for model (2.2) that*

$$\begin{aligned}
\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| &= O_P((1 + (p^{1-\delta} s^{-1})^{1/2}) h_n) \\
&= O_P(h_n + (ps^{-1})^{1/2}(p^{-\delta/2} + n^{-l_x} + n^{-l_{x\epsilon}} + p^{\delta/2} n^{-l_\epsilon})).
\end{aligned}$$

Furthermore,

$$\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = O_P(p^{1-\delta} h_n) = \|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\|$$

provided $p^{1-\delta} h_n = o(1)$.

Note that if $p \geq n$, $\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}$ is singular and the rate for the inverse sample covariance matrix becomes unbounded. If the factors are weak (i.e. $\delta > 0$), p will still be in the above rate for $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1}$. On the other hand if the factors are strong (i.e. $\delta = 0$) and $s \asymp p$, the rate is independent of p. Hence the factor-based estimator for the precision matrix $\mathbf{\Sigma}_y^{-1}$ is consistent in spectral norm irrespective of the dimension of the problem. Note that the condition $s \asymp p$ is fulfilled when the number of groups with different error variances is small. In this case, the above rate is better than the Frobenius norm convergence rate obtained in Theorem 3 of Fan et al. (2008). This is not surprising since the spectral norm is always smaller than the Frobenius norm. On the other hand, the sample precision matrix has the convergence rate linear in p, which is the same as for the sample covariance matrix.

Fan et al. (2008) studied the rate of convergence for a factor model-based precision matrix estimator under the Frobenius norm and the transformed Frobenius norm $\|\cdot\|_{\mathbf{\Sigma}}$ defined as

$$\|\mathbf{B}\|_{\mathbf{\Sigma}} = p^{-1/2}\|\mathbf{\Sigma}^{-1/2}\mathbf{B}\mathbf{\Sigma}^{-1/2}\|_F,$$

where $\|\cdot\|_F$ denotes the Frobenius norm. They show that the convergence rate under the Frobenius norm still depends on p, while the rate under the transformed Frobenius norm $\mathbf{\Sigma} = \mathbf{\Sigma}_{\mathbf{y}}$ is independent of p. Since $\mathbf{\Sigma}_{\mathbf{y}}$ is unknown in practice, the latter result has little practical impact.

4 Combination of Strong and Weak Stationary Factors

In this section we present the convergence results for stationary factors with different strength. It is an important generalization to model (2.2) since we have seen from Theorem 1 and 3 that weak factors worsen the rates of convergence for our estimators. It turns out that weak factors can worsen the rates of convergence even at the presence of strong factors, as will be shown in Theorem 4 and 7.

Like section 2.1, we introduce the factor model as

$$\mathbf{y}_t = \mathbf{A}_1\mathbf{x}_{1t} + \mathbf{A}_2\mathbf{x}_{2t} + \boldsymbol{\epsilon}_t, \tag{4.8}$$

where for $j = 1, 2$, $\mathbf{x}_{jt}$ is a $r_j \times 1$ times series of unobserved factors. $\mathbf{A}_j$ is a $p \times r_j$ constant factor loadings matrix, and $r_j \leq p$ is the number of factors (assumed known) for the j-th group of factors, with the grouping defined in assumption (B)' to follow. The white noise process $\{\boldsymbol{\epsilon}_t\}$ has mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$. Assumptions (I) to (III) in section 2.1 are to be satisfied by both factors series $\{\mathbf{x}_{1t}\}$ and $\{\mathbf{x}_{2t}\}$.

We define $\boldsymbol{\Sigma}_{ij}(k) = \mathrm{Cov}(\mathbf{x}_{i,t+k}, \mathbf{x}_{j,t})$ and $\boldsymbol{\Sigma}_{i\boldsymbol{\epsilon}}(k) = \mathrm{Cov}(\mathbf{x}_{i,t+k}, \boldsymbol{\epsilon}_t)$. Also, $\widetilde{\boldsymbol{\Sigma}}_{ij}(k)$ and $\widetilde{\boldsymbol{\Sigma}}_{i\boldsymbol{\epsilon}}(k)$ are the respective sample version of them, defined similar to those in section 2.2. We need some regularity conditions similar to assumptions (A) to (C) in section 2.2:

(A)' The factor process $\{\mathbf{x}_{jt}\}$ is stationary for $j = 1, 2$. For $k = 0, 1, \cdots, k_0$, where k_0 is a small positive integer, we have for $i, j = 1, 2$,

$$\|\boldsymbol{\Sigma}_{ij}(k)\| \asymp 1 \asymp \|\boldsymbol{\Sigma}_{ij}(k)\|_{\min}.$$

The cross autocovariance matrix $\boldsymbol{\Sigma}_{i\boldsymbol{\epsilon}}(k)$ has elements of order $O(1)$.

(B)' For $j = 1, 2$, $\mathbf{A}_j = (\mathbf{a}_{j1} \cdots \mathbf{a}_{jr_j})$ such that $\|\mathbf{a}_{ji}\|^2 \asymp p^{1-\delta_j}$, $\ i = 1, \cdots, r_j$, $\ 0 \le \delta_j \le 1$. Assume WLOG $\delta_1 < \delta_2$.

(C)' For each $j = 1, 2$ and $i = 1, \cdots, r_j$ and δ_j given in (B)',

$$\min_{\theta_{1k}, \theta_{2k}} \|\mathbf{a}_{1i} - \sum_{k \ne i} \theta_{1k}\mathbf{a}_{1k} - \sum_{k} \theta_{2k}\mathbf{a}_{2k}\|^2 \asymp p^{1-\delta_1},$$
$$\min_{\theta_{2k}} \|\mathbf{a}_{2i} - \sum_{k \ne i} \theta_{2k}\mathbf{a}_{2k}\|^2 \asymp p^{1-\delta_2}.$$

Assumptions (A)' is similar to (A) in section 2.2. Assumption (B)' groups the factors into two groups, with the first group being the stronger factors and the second group the weaker factors (since $\delta_1 < \delta_2$). Assumption (C)' says that both factor loadings matrices $\mathbf{A}_1$ and $\mathbf{A}_2$ do not have columns close enough to any linear combinations of others. Otherwise there should be other weaker factors than those in $\{\mathbf{x}_{1t}\}$ or $\{\mathbf{x}_{2t}\}$.

With these assumptions, we can write the model in a form similar to model (2.2):

$$\begin{aligned}
&\mathbf{y}_t = \mathbf{A}_1\mathbf{x}_{1t} + \mathbf{A}_2\mathbf{x}_{2t} + \boldsymbol{\epsilon}_t, \ \ \mathbf{A}_j^T\mathbf{A}_j = \mathbf{I}_{r_j} \text{ and } \mathbf{A}_1^T\mathbf{A}_2 = \mathbf{0}, \text{ with} \\
&\|\boldsymbol{\Sigma}_{jj}(k)\| \asymp p^{1-\delta_j} \asymp \|\boldsymbol{\Sigma}_{jj}(k)\|_{\min}, \ \ \|\boldsymbol{\Sigma}_{12}(k)\| \asymp p^{1-\delta_1/2-\delta_2/2} \asymp \|\boldsymbol{\Sigma}_{12}(k)\|_{\min}, \\
&\|\boldsymbol{\Sigma}_{j\boldsymbol{\epsilon}}(k)\| = O(p^{1-\delta_j/2}), \qquad (4.9) \\
&\mathrm{Cov}(\mathbf{x}_{js}, \boldsymbol{\epsilon}_t) = \mathbf{0} \text{ for all } s \le t, \\
&\mathrm{Cov}(\boldsymbol{\epsilon}_s, \boldsymbol{\epsilon}_t) = \mathbf{0} \text{ for all } s \ne t.
\end{aligned}$$

The derivation from model (4.8) using assumptions (A)' to (C)' is similar to that in section 7 from model (2.1) to (2.2) using assumptions (A) to (C), and is thus omitted.

4.1 Simple vs two-step procedures for $\mathbf{A}$

Peña and Poncela (2006) has considered two procedures for estimating $\mathbf{A}$, with the two-step procedure used when some of the factors are too weak, resulting in very small eigenvalues which cannot be distinguished from zero eigenvalues in practice. We analyze these two procedures, and give more insights on their respective advantages in Theorem 4 and 7, when the dimension of the data p grows with n and some of the factors are not necessarily strong.

The simple procedure is exactly the same as the estimation procedure in section 2.3, where now $\mathbf{A} = (\mathbf{A}_1\ \mathbf{A}_2)$, and $\mathbf{x}_t = (\mathbf{x}_{1t}^T\ \mathbf{x}_{2t}^T)^T$. And similar to section 2.3, we assume that the factor loadings matrix $\mathbf{A}$ and factors $\mathbf{x}_t$ are rotated with $\mathbf{Q}_x$, and model (4.9) is the model obtained after such rotation. The estimated factors and residuals have the same formulae as in (2.6).

The two-step procedure, on the other hand, remove the effects from the stronger factors first and then estimate the corresponding factor loadings using the procedure in section 2.3. Assuming r_1 is known (see the remark at the end of this section), we remove the effect of the stronger factors by evaluating

$$\mathbf{y}_t^* = \mathbf{y}_t - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T\mathbf{y}_t, \tag{4.10}$$

where $\widehat{\mathbf{A}}_1$ is a $p \times r_1$ matrix, the estimator of $\mathbf{A}_1$. Then we treat $\mathbf{y}_t^*$ as the raw data, and calculate the $p \times r_2$ matrix $\widehat{\mathbf{A}}_2$ using the simple procedure as in section 2.3.

4.2 Theoretical results for the two procedures

To present the theoretical properties for the procedures, we need to introduce an assumption similar to assumption (D) in section 3:

(D)' It holds for any $0 \le k \le k_0$ that for $i = 1, 2$, the elementwise rates of convergence for $\widetilde{\mathbf{\Sigma}}_{ii}(k) - \mathbf{\Sigma}_{ii}(k)$, $\widetilde{\mathbf{\Sigma}}_{12}(k) - \mathbf{\Sigma}_{12}(k)$, $\widetilde{\mathbf{\Sigma}}_{i\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{i\boldsymbol{\epsilon}}(k)$ and $\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}(k)$ are respectively $O_P(n^{-l_i})$, $O_P(n^{-l_{12}})$, $O_P(n^{-l_{i\epsilon}})$ and $O_P(n^{-l_\epsilon})$, for some constants $0 \le l_i, l_{12}, l_{i\epsilon}, l_\epsilon \le 1/2$. We also have, elementwise, $\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}i}(k) = O_P(n^{-l_{i\epsilon}})$.

Theorem 4 *Under model (4.9) and assumption (D)', if* $\|\mathbf{\Sigma}_{i\boldsymbol{\epsilon}}(k)\| = o(\|\mathbf{\Sigma}_{ii}(k)\|)$ *for* $i = 1, 2$, *the simple procedure yields*

$$\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(\omega_1), \quad \|\widehat{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(\omega_2) = \|\widehat{\mathbf{A}} - \mathbf{A}\|,$$

where $\mathbf{A} = (\mathbf{A}_1\ \mathbf{A}_2)$*, and*

$$\omega_1 = n^{-l_1} + p^{\delta_1-\delta_2}n^{-l_2} + p^{\frac{\delta_1-\delta_2}{2}}n^{-l_{12}} + p^{\delta_1/2}n^{-l_{1\epsilon}} + p^{\delta_1-\delta_2/2}n^{-l_{2\epsilon}} + p^{\delta_1}n^{-l_\epsilon},$$

$$\omega_2 = \begin{cases} p^{2\delta_2-2\delta_1}\omega_1, & \text{if } \|\boldsymbol{\Sigma}_{21}(k)\|_{\min} = o(p^{1-\delta_2}); \\ p^{c-2\delta_1}, & \text{if } \|\boldsymbol{\Sigma}_{21}(k)\|_{\min} \asymp p^{1-c/2}, \text{ with } \delta_1+\delta_2 < c < 2\delta_2; \\ p^{\delta_2-\delta_1}\omega_1, & \text{if } \|\boldsymbol{\Sigma}_{21}(k)\| \asymp p^{1-\delta_1/2-\delta_2/2} \asymp \|\boldsymbol{\Sigma}_{21}(k)\|_{\min}, \end{cases}$$

with the l's as defined in assumption (D)', and that we assume $\omega_1, \omega_2 = o(1)$. The two step procedure yields, for some rotation $\mathbf{A}_2'$ of $\mathbf{A}_2$,

$$\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(\omega_1), \quad \|\widehat{\mathbf{A}}_2' - \mathbf{A}_2'\| = O_P(p^{\delta_2-\delta_1}\omega_1) = \|\widehat{\mathbf{A}} - \mathbf{A}\|.$$

From this theorem, when one factor is stronger than the other one, it can be better to estimate the factor loadings matrix of the stronger factors and remove its effects before estimating the weaker factors, especially when the stronger and weaker factors are not having strong correlations with each other. See section 4.1.

In practice, we can inspect the magnitudes of the elements of $\widehat{\mathbf{x}}_t$ and see which represent stronger and which represent weaker factors, hence getting an idea on the value of r_1 for performing the two-step procedure. This practice is supported by the following theorem.

Theorem 5 *Under conditions of Theorem 4, provided $p^{\delta_2-\delta_1}\omega_1 = o(1)$, the simple procedure has $\|\widehat{\mathbf{x}}_{2t}\| = o_P(\|\widehat{\mathbf{x}}_{1t}\|)$.*

In general, the above theorem tells us that we can group the factors by looking at the magnitudes of the estimated factor series. Then we can calculate the corresponding eigenvectors for the group with the largest order of magnitude, estimate the corresponding factors, and then remove the effect from this group using formula (4.10). We then treat $\mathbf{y}_t^*$ as the raw data again and repeat the process until all the factors are found. This should give us better estimates for the weaker factors if not all the weaker factors are having strong correlations with the stronger factors.

The next two theorems show the rates for estimating the covariance and the precision matrices using the two procedures, which need a modified version of assumption (M2):

(M2)' From assumption (M1), define $s = \min_{1 \le i \le k} m_i$. Under model (4.9), let $\widehat{\mathbf{A}}$ be the estimator from Theorem 4. Then

$$s^{-1} < p^{1-\delta_1}\|\widehat{\mathbf{A}} - \mathbf{A}\|^2 \ \text{ and } \ (1 + (p^{1-\delta_1}s^{-1})^{1/2})\|\widehat{\mathbf{A}} - \mathbf{A}\| = o_P(1).$$

Theorem 6 *Under model (4.9) and assumption (D)', the sample covariance matrix has*

$$\begin{aligned}\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| &= O_P(p^{1-\delta_1}n^{-l_1} + p^{1-\delta_2}n^{-l_2} + p^{1-\delta_1/2}n^{-l_{1\epsilon}} + p^{1-\delta_2/2}n^{-l_{2\epsilon}} + pn^{-l_\epsilon}) \\ &= O_P(p^{1-\delta_1}\omega_1), \\ \|\widehat{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| &= O_P(p^{1-\delta_1}\|\widehat{\mathbf{A}} - \mathbf{A}\|),\end{aligned}$$

where $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}$ is the covariance matrix estimator defined by (3.7) with $\widehat{\mathbf{A}}$ either obtained by the simple or the two-step procedure, and the rates in Theorem 4 apply.

This theorem corresponds to Theorem 2, which says that the factor model-based covariance matrix estimator cannot perform better than the sample counterpart in terms of rate of convergence in spectral norm. Indeed the factor model-based estimator can do worse than the sample covariance matrix under model (4.9), especially when the simple procedure is used when the factors are in fact of very different strength, rendering a worse rate for $\|\widehat{\mathbf{A}} - \mathbf{A}\|$.

Theorem 7 *For model (4.9), under assumptions (D)', (M1) in section 3 and (M2)', the precision matrix estimator $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1}$ has*

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = O_P((1 + (p^{1-\delta_1}s^{-1})^{1/2})\|\widehat{\mathbf{A}} - \mathbf{A}\|),$$

where $\widehat{\mathbf{A}}$ is either obtained by the simple or the two-step procedure, and the rates in Theorem 4 apply. The inverse sample covariance matrix $\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}^{-1}$, if $\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| = o_P(1)$, has

$$\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = O_P(\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\|).$$

As in Theorem 3, the inverse covariance matrix improves over the sample counterpart, especially when the two-step procedure is used when factors are of different strength.

5 Simulations

In this section, we demonstrate how dimensionality affects our estimators, and illustrate finite sample performance through simulations, while the simple and two-step procedures are compared. Analysis of a set of implied volatility data using our factor model and methodology is presented in the next section.

Example 1

We start by a simple toy example to illustrate the results in Theorem 1, 2 and 3. Assume a one factor model

$$\mathbf{y}_t = \mathbf{A}x_t + \boldsymbol{\epsilon}_t, \;\; \epsilon_{tj} \sim \text{ i.i.d. } N(0, 2^2),$$

where the factor loadings and the factor time series are respectively $(\mathbf{A})_i = 2\cos(2\pi i/p)$ and $x_t = 0.9x_{t-1} + \eta_t$, $i = 1, \cdots, p$, $t = 1, \cdots, n$, with $\eta_t \sim N(0, 2^2)$ and is independent of all other variables. Hence we have a strong factor for this model. We set $n = 200, 500$ and $p = 20, 180, 400, 1000$, and for each (n, p) combination we simulate from the model 50 times and calculate the estimation errors as in Theorems 1, 2 and 3.

$n = 200$	$\Vert\widehat{\mathbf{A}} - \mathbf{A}_x\Vert$	$\Vert\mathbf{S}^{-1} - \boldsymbol{\Sigma}_{\mathbf{y}}^{-1}\Vert$	$\Vert\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}^{-1} - \boldsymbol{\Sigma}_{\mathbf{y}}^{-1}\Vert$	$\Vert\mathbf{S} - \boldsymbol{\Sigma}_{\mathbf{y}}\Vert$	$\Vert\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}} - \boldsymbol{\Sigma}_{\mathbf{y}}\Vert$
$p = 20$	$.022_{(.005)}$	$.24_{(.03)}$	$.009_{(.002)}$	$218_{(165)}$	$218_{(165)}$
$p = 180$	$.023_{(.004)}$	$79.8_{(29.8)}$	$.007_{(.001)}$	$1962_{(1500)}$	$1963_{(1500)}$
$p = 400$	$.022_{(.004)}$	-	$.007_{(.001)}$	$4102_{(3472)}$	$4103_{(3471)}$
$p = 1000$	$.023_{(.004)}$	-	$.007_{(.001)}$	$10797_{(6820)}$	$10800_{(6818)}$

Table 1: *Mean of estimation errors for the one factor example.* $\mathbf{S}$ *is the sample covariance matrix. The numbers in brackets are the corresponding standard deviations.*

The results when $n = 500$ is similar and thus not displayed. It is clear from Table 1 that the estimation errors in L_2 norm for $\widehat{\mathbf{A}}$ and $\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}^{-1}$ are independent of p, which agree with the results from Theorem 1 and Theorem 3 with $\delta = 0$, as we have a strong factor in this example. The inverse of the sample covariance matrix $\mathbf{S}^{-1}$ is not defined for $p > n$, and is a bad estimator even when $p < n$ as seen in above table. The last two columns show the results for $\mathbf{S}$ and $\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}$, which agree with Theorem 2 that the rates are increasing with p.

Example 2

Next, we demonstrate the effect of dimensionality when strength of factors can be different. We generate our data from model (4.9), where we generate three stationary factors

$$\begin{aligned}
x_{1,t} &= -0.8x_{1,t-1} + 0.9e_{1,t-1} + e_{1,t}, \\
x_{2,t} &= -0.7x_{2,t-1} + 0.85e_{2,t-1} + e_{2,t}, \\
x_{3,t} &= \;\;\; 0.8x_{2,t} - 0.5x_{3,t-1} + e_{3,t},
\end{aligned}$$

with $e_{i,t} \sim i.i.d.\ N(0, 1)$ for all i and t. We consider sample size $n = 100, 200, 500, 1000$ and dimension of vector $\mathbf{y}_t$ being $p = 100, 200, 500$. One of the factors has strength δ_1

and the other two have strength δ_2, both have values $0, 0.5$ or 1. For each combination of $(n, p, \delta_1, \delta_2)$, we simulate $N = 100$ times and calculate the mean and SD of the error measures for comparison.

For each column of $\mathbf{A}$, we generate the first $p/2$ elements randomly from the $U(-2, 2)$ distribution; the rest are set to zero. This can imitate real situations where some cross-sectional observations are just noise, and can give extra difficulty in identifying factor structures. We then adjust the strength of the factors by normalizing the columns, setting $\mathbf{a}_i/p^{\delta_i/2}$ as the i-th column of $\mathbf{A}$ (we set $\delta_2 = \delta_3$).

Finally, we generate the $\boldsymbol{\epsilon}_t$ with $\epsilon_{j,t} \sim i.i.d.\ N(0, \sigma_j^2)$ where $\sigma_j = 0.5$ if j is odd, and $\sigma_j = 0.8$ otherwise. Since our theories have not assumed on the distribution of the noise term $\boldsymbol{\epsilon}_t$ other than $\mathbf{\Sigma}_{\boldsymbol{\epsilon}}$ itself, we demonstrate the effect of heavy-tailed noise on our estimators by generating $\epsilon_{j,t} \sim i.i.d.\ t_5$ (suitably normalized to make σ_j^2 the same as those in the normal simulations) in another set of simulations.

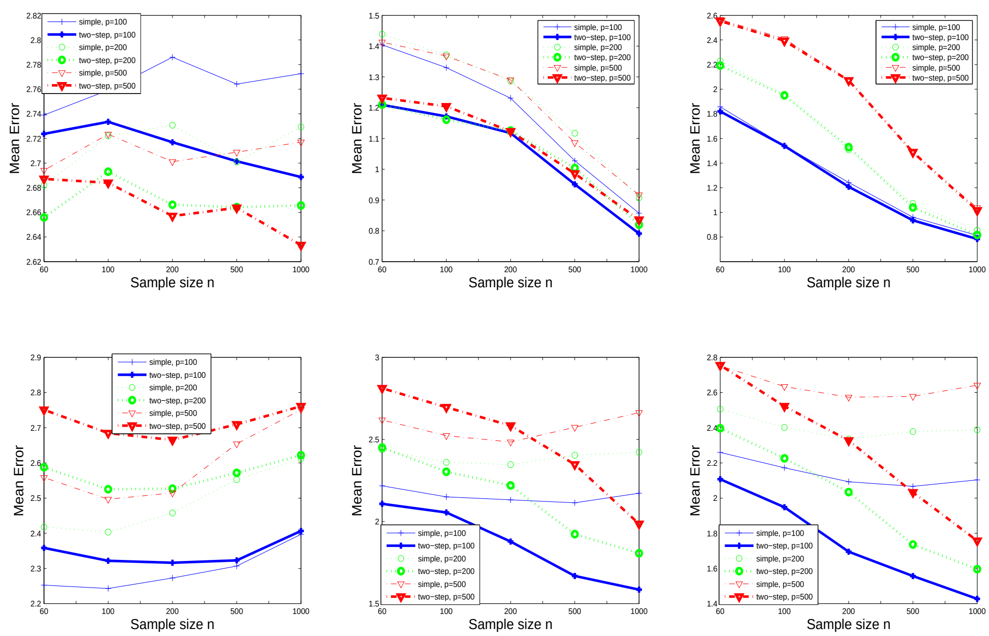

Figure 1: *Mean of* $\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\|$*, noises distributed as* t_5*. Top row:* $\delta_1 = \delta_2 = \delta_3 = 0$*. Left: Two factors; Middle: Three factors; Right: Four factors. Bottom row: Same, but* $\delta_1 = 0, \delta_2 = \delta_3 = 0.5$*.*

For the covariance matrix estimation, our results (not shown) show that, as in Theorem 6, both the sample covariance matrix and factor model based one are poor estimators when p is large. In fact both the simple and two-step procedures yield worse estimation errors than the sample covariance matrix, although performance gap closes down as n gets larger.

For precision matrix estimation, figure 1 shows clearly that when the number of factors are not underestimated, the two step procedure outperforms the simple one when strength of factors are different, and performs at least as good when the factors are of the same strength. The performance is better when the number of factors is in fact more than the optimal because we have used $k_0 = 3$ instead of just 1 or 2 when the serial correlations for the factors are in fact quite weak. Hence we accumulate pure noises, which sometimes introduces non-genuine factors that are stronger than the genuine ones, and requires the inclusion of more than necessary factors to reduce the errors. Not shown here, we have repeated the simulations with $k_0 = 1$, and the performance is much better and is optimal when the number of factors is 3. The two-step procedure still outperform the simple one. The simulations with normal errors are not shown here since the results are similar.

6 Data Analysis : Implied Volatility Surfaces

We illustrate the methodology developed through modeling the dynamic behavior of IBM, Microsoft and Dell implied volatility surfaces. The data was obtained from OptionMetrics via the WRDS database. The dates in question are 03/01/2006 – 29/12/2006 (250 days in total). For each day t we observe the implied volatility $W_t(u_i, v_j)$ computed from call options as a function of time to maturity of 30, 60, 91, 122, 152, 182, 273, 365, 547 and 730 calender days which we denote by u_i, $i = 1, \ldots, p_u$ ($p_u = 10$) and deltas of 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, and 0.8 which we denote by v_j, $j = 1, \ldots, p_v$ ($p_v = 13$). We collect these implied volatilities in the matrix $\mathbf{W}_t = (W_t(u_i, v_i)) \in \mathbb{R}^{p_u \times p_v}$. Figure 2 displays the mean volatility surface of IBM, Microsoft and Dell over the period in question. It is clear from this graphic that the implied volatilities surfaces are not flat. Indeed any cross-section in the maturity or delta axis display the well documented volatility smile.

It is a well documented stylized fact that implied volatilities are non-stationary (see Cont and da Fonseca (1988), Fengler et al. (2007) and Park et al. (2009) amongst others). Indeed, when applying the Dickey-Fuller test to each of the univariate time series

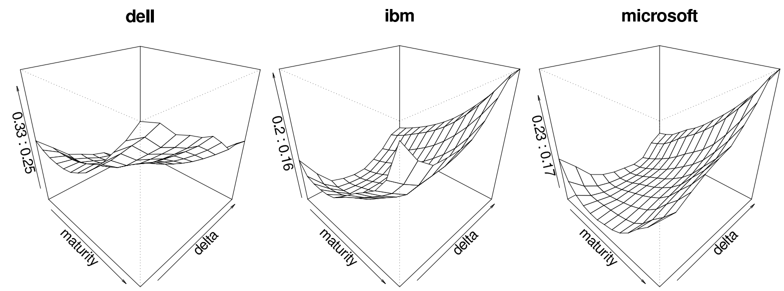

Figure 2: *Mean implied volatility surfaces.*

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$W_t(u_i, v_i)$, none of the $p_u \times p_v = 130$ nulls of unit roots could be rejected at the 10% level. Of course we should treat the results of these tests with some caution since we are performing a large number of hypothesis tests, but even still the evidence in favor of unit roots is overwhelming. Therefore, instead of working with $\mathbf{W}_t$ directly, we choose to work with $\Delta \mathbf{W}_t = \mathbf{W}_t - \mathbf{W}_{t-1}$. Our observations are then $\mathbf{y}_t = \text{vec}\{\Delta \mathbf{W}_t\}$, where for any matrix $\mathbf{M} = (\mathbf{m}_1, \ldots, \mathbf{m}_{p_v}) \in \mathbb{R}^{p_u \times p_v}$, $\text{vec}\{\mathbf{M}\} = (\mathbf{m}_1^T, \ldots, \mathbf{m}_{p_v}^T)^T \in \mathbb{R}^{p_u p_v}$. Note that $\mathbf{y}_t$ is now defined over $04/01/2006 - 29/12/2006$ since we lose an observation due to differencing. Hence altogether there are 249 time points, and the dimension of $\mathbf{y}_t$ is $p = p_v \times p_u = 130$.

We perform the factor model estimation on a rolling window of length 100 days. A window is defined from the i-th day to the $(i+99)$-th day for $i = 1, \cdots, 150$. The length of the window is chosen so that the stationary assumption of the data is approximately satisfied. For each window, we compare our methodology with the least squares based methodology by Bai and Ng (2002) by estimating the factor loadings matrix and the factors series for the two methods. For the i-th window, we use an AR model to forecast the $(i+100)$-th value of the estimated factor series $\mathbf{x}_{i+100}^{(1)}$, so as to obtain a one-step ahead forecast $\mathbf{y}_{i+100}^{(1)} = \widehat{\mathbf{A}}\mathbf{x}_{i+100}^{(1)}$ for $\mathbf{y}_{i+100}$. We then calculate the RMSE for the $(i+100)$-th day defined by

$$\text{RMSE} = p^{-1/2} \left\{ \sum_{j=1}^{p} \|\mathbf{y}_{i+100}^{(1)} - \mathbf{y}_{i+100}\|^2 \right\}^{1/2}.$$

More in depth theoretical as well as data analysis for forecasting is given in Lam et al. (2010).

6.1 Estimation results

In forming the matrix $\widetilde{\mathbf{L}}$ for each window, we take $k_0 = 5$ in (2.5) , taking advantage that the autocorrelations are not weak even at higher lags, though similar results (not reported here) are obtained for smaller k_0.

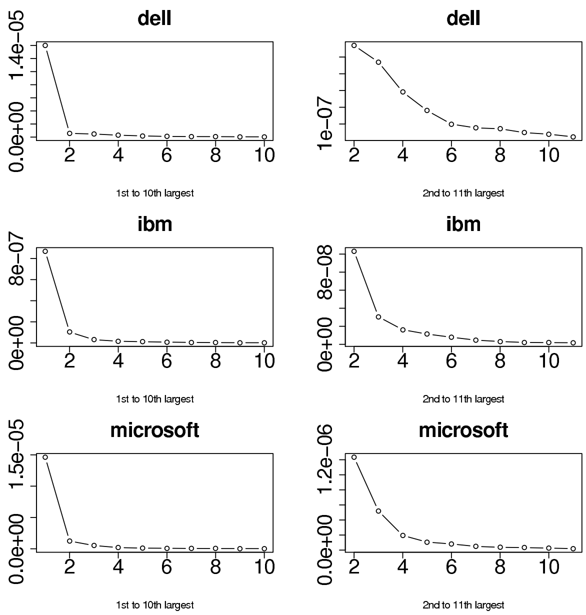

Figure 3: *Average of each ordered eigenvalue of* $\widetilde{\mathbf{L}}$ *over the 150 windows. Left: Ten largest. Right: Second to eleventh largest.*

Figure 3 displays the average of each ordered eigenvalue over the 150 windows. The left hand side shows the average of the largest to the average of the tenth largest eigenvalue of $\widetilde{\mathbf{L}}$ for Dell, IBM and Microsoft for our method, whereas the right hand side shows the second to eleventh largest. We obtain similar results for the Bai and Ng (2002) procedure and thus the corresponding graph is not shown.

From this graphic it is apparent that there is one eigenvalue that is much larger than the others for all three companies for each window. We have done automatic selection for the number of factors for each window using the IC_{p1} criterion in Bai and Ng (2002) and a one factor model is consistently obtained for each window and for each company. Hence both methods chose a one factor model over the 150 windows.

Figure 4 displays the cumulative RMSE over the 150 windows for each method. We choose a benchmark procedure (green line in each plot), where we just treat today's value as the one-step ahead forecast. Except for Dell where Bai and Ng (2002) procedure

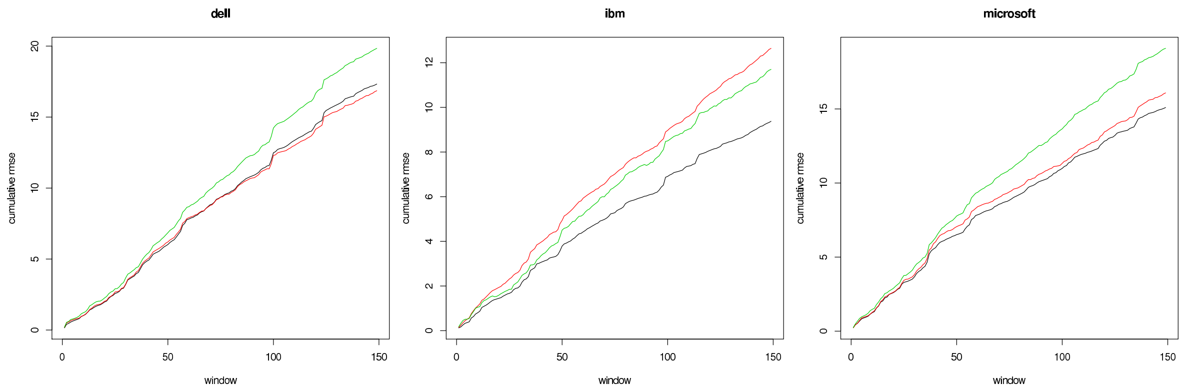

Figure 4: *The cumulative RMSE over the 150 windows. Red: Bai and Ng (2002) procedure. Green: Taking forecast* $\mathbf{y}_{t+1}^{(1)}$ *to be* $\mathbf{y}_t$*. Black: Our methodology.*

is doing marginally better, our methodology consistently outperforms the benchmark procedure and is better than Bai and Ng (2002) for IBM and Microsoft.

7 Proofs

First of all, we show how model (2.2) can be derived from (2.1).

Applying the standard QR decomposition, we may write $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ is a $p \times r$ matrix such that $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_r$, $\mathbf{R}$ is an $r \times r$ upper triangular matrix. Therefore model (2.1) can be expressed as

$$\mathbf{y}_t = \mathbf{Q}\mathbf{x}_t' + \boldsymbol{\epsilon}_t,$$

where $\mathbf{x}_t' = \mathbf{R}\mathbf{x}_t$. With assumptions (A) to (C), the diagonal entries of $\mathbf{R}$ are all asymptotic to $p^{\frac{1-\delta}{2}}$. Since r is a constant, using

$$\|\mathbf{R}\| = \max_{\|\mathbf{u}\|=1} \|\mathbf{R}\mathbf{u}\|, \quad \|\mathbf{R}\|_{\min} = \min_{\|\mathbf{u}\|=1} \|\mathbf{R}\mathbf{u}\|,$$

and the fact that $\mathbf{R}$ is an $r \times r$ upper triangular matrix with all diagonal elements having the largest order $p^{\frac{1-\delta}{2}}$, we have

$$\|\mathbf{R}\| \asymp p^{\frac{1-\delta}{2}} \asymp \|\mathbf{R}\|_{\min}.$$

Thus, for $k = 1, \cdots, k_0$, $\boldsymbol{\Sigma}_{\mathbf{x}'}(k) = \text{Cov}(\mathbf{x}_{t-k}', \mathbf{x}_t') = \mathbf{R}\boldsymbol{\Sigma}_{\mathbf{x}}(k)\mathbf{R}^T$, with

$$p^{1-\delta} \asymp \|\mathbf{R}\|_{\min}^2 \cdot \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|_{\min} \leq \|\boldsymbol{\Sigma}_{\mathbf{x}'}(k)\|_{\min} \leq \|\boldsymbol{\Sigma}_{\mathbf{x}'}(k)\| \leq \|\mathbf{R}\|^2 \cdot \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\| \asymp p^{1-\delta},$$

so that $\|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\| \asymp p^{1-\delta} \asymp \|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|_{\min}$. We used $\|\mathbf{AB}\|_{\min} \geq \|\mathbf{A}\|_{\min} \cdot \|\mathbf{B}\|_{\min}$, which can be proved by noting

$$\begin{aligned}\|\mathbf{AB}\|_{\min} = \min_{\mathbf{u}\neq\mathbf{0}} \frac{\mathbf{u}^T\mathbf{B}^T\mathbf{A}^T\mathbf{ABu}}{\|\mathbf{u}\|^2} &\geq \min_{\mathbf{u}\neq\mathbf{0}} \frac{(\mathbf{Bu})^T\mathbf{A}^T\mathbf{A}(\mathbf{Bu})}{\|\mathbf{Bu}\|^2} \cdot \frac{\|\mathbf{Bu}\|^2}{\|\mathbf{u}\|^2} \\ &\geq \min_{\mathbf{w}\neq\mathbf{0}} \frac{\mathbf{w}^T\mathbf{A}^T\mathbf{Aw}}{\|\mathbf{w}\|^2} \cdot \min_{\mathbf{u}\neq\mathbf{0}} \frac{\|\mathbf{Bu}\|^2}{\|\mathbf{u}\|^2} = \|\mathbf{A}\|_{\min} \cdot \|\mathbf{B}\|_{\min}. \qquad (7.1)\end{aligned}$$

Finally, using assumption (A) that $\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k) = O(1)$ elementwise, and that it has $rp \asymp p$ elements, we have

$$\|\boldsymbol{\Sigma}_{\mathbf{x}',\boldsymbol{\epsilon}}(k)\| = \|\mathbf{R}\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| \leq \|\mathbf{R}\| \cdot \|\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\|_F = O(p^{\frac{1-\delta}{2}}) \cdot O(p^{1/2}) = O(p^{1-\delta/2}).$$

Before proving the theorems in section 3, we need to have three lemmas.

Lemma 1 *Under the factor model (2.2) which is a reformulation of (2.1), and under condition (D) in section 2.2, we have for* $0 \leq k \leq k_0$,

$$\begin{gathered}\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k) - \boldsymbol{\Sigma}_{\mathbf{x}}(k)\| = O_P(p^{1-\delta}n^{-l_x}), \qquad \|\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}(k)\| = O_P(pn^{-l_\epsilon}), \\ \|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}}(k) - \boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| = O_P(p^{1-\delta/2}n^{-l_{x\epsilon}}) = \|\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon},\mathbf{x}}(k) - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon},\mathbf{x}}(k)\|,\end{gathered}$$

for some constants $0 < l_x, l_{x\epsilon}, l_\epsilon \leq 1/2$. *Moreover,* $\|\mathbf{x}_t\|^2 = O_P(p^{1-\delta})$ *for all real* t.

Proof. Using the notations in section 2.2, let $\mathbf{x}_t$ be the factors in model (2.1), and $\mathbf{x}'_t$ be the factors in model (2.2), with the relation that $\mathbf{x}'_t = \mathbf{R}\mathbf{x}_t$, where $\mathbf{R}$ is an upper triangular matrix with $\|\mathbf{R}\| \asymp p^{\frac{1-\delta}{2}} \asymp \|\mathbf{R}\|_{\min}$ (see the start of this section for more details on $\mathbf{R}$). Then we immediately have $\|\mathbf{x}'_t\|^2 \leq \|\mathbf{R}\|^2 \cdot \|\mathbf{x}_t\|^2 = O_P(p^{1-\delta}r) = O_P(p^{1-\delta})$.

Also, the covariance matrix and the sample covariance matrix for $\{\mathbf{x}'_t\}$ are respectively

$$\boldsymbol{\Sigma}_{\mathbf{x}'}(k) = \mathbf{R}\boldsymbol{\Sigma}_{\mathbf{x}}(k)\mathbf{R}^T, \quad \widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}'}(k) = \mathbf{R}\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k)\mathbf{R}^T,$$

where $\boldsymbol{\Sigma}_{\mathbf{x}}(k)$ and $\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k)$ are respectively the covariance matrix and the sample covariance matrix for the factors $\{\mathbf{x}_t\}$. Hence

$$\begin{aligned}\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}'}(k) - \boldsymbol{\Sigma}_{\mathbf{x}'}(k)\| &\leq \|\mathbf{R}\|^2 \cdot \|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k) - \boldsymbol{\Sigma}_{\mathbf{x}}(k)\| \\ &= O(p^{1-\delta}) \cdot O_P(n^{-l_x} \cdot r) \\ &= O_P(p^{1-\delta}n^{-l_x}),\end{aligned}$$

which is the rate specified in the lemma. We used the fact that the matrix $\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}}(k) - \boldsymbol{\Sigma}_{\mathbf{x}}(k)$ has r^2 elements, with elementwise rate of convergence being $O(n^{-l_x})$ as in assumption (D). Other rates can be derived similarly. $\square$

The following is Theorem 8.1.10 in Golub and Van Loan (1996), which is stated explicitly since most of our main theorems are based on this. See Johnstone and Arthur (2009) also.

Lemma 2 *Suppose* $\mathbf{A}$ *and* $\mathbf{A}+\mathbf{E}$ *are* $n \times n$ *symmetric matrices and that*

$$\mathbf{Q} = [\mathbf{Q}_1 \;\; \mathbf{Q}_2] \quad (\mathbf{Q}_1 \;\; \textit{is} \;\; n \times r, \; \mathbf{Q}_2 \;\; \textit{is} \;\; n \times (n-r))$$

is an orthogonal matrix such that $\mathrm{span}(\mathbf{Q}_1)$ *is an invariant subspace for* $\mathbf{A}$ *(i.e.,* $\mathrm{span}(\mathbf{Q}_1) \subset \mathrm{span}(\mathbf{A})$*). Partition the matrices* $\mathbf{Q}^T\mathbf{A}\mathbf{Q}$ *and* $\mathbf{Q}^T\mathbf{E}\mathbf{Q}$ *as follows:*

$$\mathbf{Q}^T\mathbf{A}\mathbf{Q} = \begin{pmatrix} \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2 \end{pmatrix} \qquad \mathbf{Q}^T\mathbf{E}\mathbf{Q} = \begin{pmatrix} \mathbf{E}_{11} & \mathbf{E}_{21}^T \\ \mathbf{E}_{21} & \mathbf{E}_{22} \end{pmatrix}.$$

If $\mathrm{sep}(\mathbf{D}_1, \mathbf{D}_2) := \min_{\lambda \in \lambda(\mathbf{D}_1),\, \mu \in \lambda(\mathbf{D}_2)} |\lambda - \mu| > 0$*, where* $\lambda(M)$ *denotes the set of eigenvalues of the matrix* M*, and*

$$\|\mathbf{E}\| \le \frac{\mathrm{sep}(\mathbf{D}_1, \mathbf{D}_2)}{5},$$

then there exists a matrix $\mathbf{P} \in \mathbb{R}^{(n-r)\times r}$ *with*

$$\|\mathbf{P}\| \le \frac{4}{\mathrm{sep}(\mathbf{D}_1, \mathbf{D}_2)} \|\mathbf{E}_{21}\|$$

such that the columns of $\widehat{\mathbf{Q}}_1 = (\mathbf{Q}_1 + \mathbf{Q}_2\mathbf{P})(\mathbf{I} + \mathbf{P}^T\mathbf{P})^{-1/2}$ *define an orthonormal basis for a subspace that is invariant for* $\mathbf{A}+\mathbf{E}$*.*

Proof of Theorem 1. Under model (2.2), the assumption that $\|\boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| = o(\|\boldsymbol{\Sigma}_{\mathbf{x}}(k)\|)$, and the definition of $\mathbf{L}$ and $\mathbf{D}_x$ in section 2.3 such that $\mathbf{L}\mathbf{A} = \mathbf{A}\mathbf{D}_x$, $\mathbf{D}_x$ has non-zero eigenvalues of order $p^{2-2\delta}$, contributed by the term $\boldsymbol{\Sigma}_{\mathbf{x}}(k)\boldsymbol{\Sigma}_{\mathbf{x}}(k)^T$. If $\mathbf{B}$ is an orthogonal complement of $\mathbf{A}$, then $\mathbf{L}\mathbf{B} = \mathbf{0}$, and

$$\begin{pmatrix} \mathbf{A}^T \\ \mathbf{B}^T \end{pmatrix} \mathbf{L}(\mathbf{A} \;\; \mathbf{B}) = \begin{pmatrix} \mathbf{D}_x & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \tag{7.2}$$

with $\mathrm{sep}(\mathbf{D}_x, \mathbf{0}) = \lambda_{\min}(\mathbf{D}_x) \asymp p^{2-2\delta}$ (see Lemma 2 for the definition of the function sep).

Define $\mathbf{E}_{\mathbf{L}} = \widetilde{\mathbf{L}} - \mathbf{L}$, where $\widetilde{\mathbf{L}}$ is defined in (2.5). Then it is easy to see that

$$\|\mathbf{E}_{\mathbf{L}}\| \le \sum_{k=1}^{k_0} \Big\{ \|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}}(k) - \boldsymbol{\Sigma}_{\mathbf{y}}(k)\|^2 + 2\|\boldsymbol{\Sigma}_{\mathbf{y}}(k)\| \cdot \|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}}(k) - \boldsymbol{\Sigma}_{\mathbf{y}}(k)\| \Big\}. \tag{7.3}$$

Suppose we can show further that

$$\|\mathbf{E_L}\| = O_P(p^{2-2\delta}n^{-l_x} + p^{2-3\delta/2}n^{-l_{x\epsilon}} + p^{2-\delta}n^{-l_\epsilon}) = O_P(p^{2-2\delta}h_n), \tag{7.4}$$

then since $h_n = o(1)$, we have from (7.2) that

$$\|\mathbf{E_L}\| = O_P(p^{2-2\delta}h_n) \leq \text{sep}(\mathbf{D}_x, \mathbf{0})/5$$

for sufficiently large n. Hence we can apply Lemma 2 to conclude that there exists a matrix $\mathbf{P} \in \mathbb{R}^{(p-r)\times r}$ such that

$$\|\mathbf{P}\| \leq \frac{4}{\text{sep}(\mathbf{D}_x, \mathbf{0})}\|(\mathbf{E_L})_{21}\| \leq \frac{4}{\text{sep}(\mathbf{D}_x, 0)}\|\mathbf{E_L}\| = O_P(h_n),$$

and $\widehat{\mathbf{A}} = (\mathbf{A} + \mathbf{BP})(\mathbf{I} + \mathbf{P}^T\mathbf{P})^{-1/2}$ is an estimator for $\mathbf{A}$. Then we have

$$\begin{aligned}
\|\widehat{\mathbf{A}} - \mathbf{A}\| &= \|(\mathbf{A}(\mathbf{I} - (\mathbf{I} + \mathbf{P}^T\mathbf{P})^{1/2}) + \mathbf{BP})(\mathbf{I} + \mathbf{P}^T\mathbf{P})^{-1/2}\| \\
&\leq \|\mathbf{I} - (\mathbf{I} + \mathbf{P}^T\mathbf{P})^{1/2}\| + \|\mathbf{P}\| \\
&\leq 2\|\mathbf{P}\| = O_P(h_n).
\end{aligned}$$

Hence it remains to show (7.4). To this end, consider for $k \geq 1$,

$$\|\mathbf{\Sigma_y}(k)\| = \|\mathbf{A\Sigma_x}(k)\mathbf{A}^T + \mathbf{A\Sigma_{x,\epsilon}}(k)\| \leq \|\mathbf{\Sigma_x}(k)\| + \|\mathbf{\Sigma_{x,\epsilon}}(k)\| = O(p^{1-\delta}) \tag{7.5}$$

by assumptions in model (2.2) and $\|\mathbf{\Sigma_{x,\epsilon}}(k)\| = o(\|\mathbf{\Sigma_x}(k)\|)$. Finally, noting $\|\mathbf{A}\| = 1$,

$$\begin{aligned}
\|\widetilde{\mathbf{\Sigma}}_\mathbf{y}(k) - \mathbf{\Sigma_y}(k)\| \leq& \|\widetilde{\mathbf{\Sigma}}_\mathbf{x}(k) - \mathbf{\Sigma_x}(k)\| + 2\|\widetilde{\mathbf{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}(k)\| \\
&+ \|\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}(k)\| \\
=& O_P(p^{1-\delta}n^{-l_x} + p^{1-\delta/2}n^{-l_{x\epsilon}} + pn^{-l_\epsilon})
\end{aligned} \tag{7.6}$$

by Lemma 1. With (7.5) and (7.6), we can conclude from (7.3) that

$$\|\mathbf{E_L}\| = O_P(\|\mathbf{\Sigma_y}(k)\| \cdot \|\widetilde{\mathbf{\Sigma}}_\mathbf{y}(k) - \mathbf{\Sigma_y}(k)\|),$$

which is exactly the order specified in (7.4). □

Proof of Theorem 2. For the sample covariance matrix $\widetilde{\mathbf{\Sigma}}_\mathbf{y}$, note that (7.6) is applicable to the case when $k = 0$, so that

$$\|\widetilde{\mathbf{\Sigma}}_\mathbf{y} - \mathbf{\Sigma_y}\| = O_P(p^{1-\delta}n^{-l_x} + p^{1-\delta/2}n^{-l_{x\epsilon}} + pn^{-l_\epsilon}) = O_P(p^{1-\delta}h_n).$$

For $\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}$, we have

$$\begin{aligned}\|\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}} - \boldsymbol{\Sigma}_{\mathbf{y}}\| &\le \|\widehat{\mathbf{A}}\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}}\widehat{\mathbf{A}}^T - \mathbf{A}\boldsymbol{\Sigma}_{\mathbf{x}}\mathbf{A}^T\| + \|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\| \\ &= O_P(\|\widehat{\mathbf{A}} - \mathbf{A}\| \cdot \|\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}}\| + \|\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}} - \boldsymbol{\Sigma}_{\mathbf{x}}\| + \|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\|).\end{aligned} \tag{7.7}$$

We first consider $\|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\| = \max_{1\le j\le p} |\widehat{\sigma}_j^2 - \sigma_j^2| \le I_1 + I_2 + I_3$, where

$$\begin{aligned} I_1 &= n^{-1}s^{-1}\|\Delta_j(\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)\mathbf{A}\mathbf{X}\|_F^2, \\ I_2 &= |n^{-1}s^{-1}\|\Delta_j(\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)\mathbf{E}\|_F^2 - \sigma_j^2|, \\ I_3 &= 2n^{-1}s^{-1}\|\Delta_j(\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)\mathbf{A}\mathbf{X}\|_F \cdot \|\Delta_j(\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)\mathbf{E}\|_F, \end{aligned}$$

with $\mathbf{X} = (\mathbf{x}_1 \cdots \mathbf{x}_n)$, $\mathbf{E} = (\boldsymbol{\epsilon}_1 \cdots \boldsymbol{\epsilon}_n)$. We have

$$\begin{aligned} I_1 &\le n^{-1}s^{-1}\|\Delta_j(\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T)(\mathbf{A} - \widehat{\mathbf{A}})\mathbf{X}\|_F^2 \\ &\le n^{-1}s^{-1}\|\Delta_j\|^2 \cdot \|\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T\|^2 \cdot \|\mathbf{A} - \widehat{\mathbf{A}}\|^2 \cdot \|\mathbf{X}\|_F^2 \\ &= O_P(p^{1-\delta}s^{-1}h_n^2), \end{aligned} \tag{7.8}$$

where we used Theorem 1 for the rate $\|\widehat{\mathbf{A}} - \mathbf{A}\|^2 = O_P(h_n^2)$, and Lemma 1 to get $\|\mathbf{X}\|_F^2 = O_P(p^{1-\delta}n)$. Also we used $\|\Delta_j\| = 1$ and $\|\mathbf{I}_p - \widehat{\mathbf{A}}\widehat{\mathbf{A}}^T\| \le 2$. For I_2, consider

$$\begin{aligned} I_2 &\le |n^{-1}s^{-1}\|\Delta_j\mathbf{E}\|_F^2 - \sigma_j^2| + n^{-1}s^{-1}\|\Delta_j\widehat{\mathbf{A}}\widehat{\mathbf{A}}^T\mathbf{E}\|_F^2 \\ &= O_P(n^{-l_\epsilon}) + O_P(n^{-1}s^{-1}(\|\Delta_j\mathbf{A}\mathbf{A}^T\mathbf{E}\|_F^2 + \|\Delta_j\mathbf{A}(\widehat{\mathbf{A}} - \mathbf{A})^T\mathbf{E}\|_F^2)) \\ &= O_P(n^{-l_\epsilon}) + O_P(n^{-1}s^{-1}(\|\mathbf{A}^T\mathbf{E}\|_F^2 + \|(\widehat{\mathbf{A}} - \mathbf{A})^T\mathbf{E}\|_F^2)) \\ &= O_P(n^{-l_\epsilon}) + O_P(n^{-1}s^{-1}(\|\mathbf{A}^T\mathbf{E}\|_F^2)), \end{aligned} \tag{7.9}$$

where we used assumption (D) in arriving at $|n^{-1}s^{-1}\|\Delta_j\mathbf{E}\|_F^2 - \sigma_j^2| = o_P(n^{-l_\epsilon})$, and that $\|(\widehat{\mathbf{A}} - \mathbf{A})^T\mathbf{E}\|_F \le \|\mathbf{A}^T\mathbf{E}\|_F$ for sufficiently large n since $\|\widehat{\mathbf{A}} - \mathbf{A}\| = o_P(1)$. Consider $\mathbf{a}_i^T\boldsymbol{\epsilon}_j$ which is the (i,j)-th element of $\mathbf{A}^T\mathbf{E}$. We have

$$E(\mathbf{a}_i^T\boldsymbol{\epsilon}_j) = 0, \quad \mathrm{Var}(\mathbf{a}_i^T\boldsymbol{\epsilon}_j) = \mathbf{a}_i^T\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\mathbf{a}_i \le \max_{1\le j\le p} \sigma_j^2 = O(1)$$

since the σ_j^2's are uniformly bounded away from infinity by assumption (M1). Hence each element in $\mathbf{A}^T\mathbf{E}$ is $O_P(1)$, which implies that

$$n^{-1}s^{-1}\|\mathbf{A}^T\mathbf{E}\|_F^2 = O_P(n^{-1}s^{-1} \cdot rn) = O_P(s^{-1}).$$

Hence from (7.9) we have

$$I_2 = O_P(n^{-l_\epsilon} + s^{-1}). \tag{7.10}$$

Assumption (M2) ensures that both I_1 and I_2 are $o_P(1)$ from (7.8) and (7.10) respectively. From these we can see that $I_3 = O_P(I_1^{1/2}) = O_P((p^{1-\delta}s^{-1})^{1/2}h_n)$, which shows that

$$\|\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}\| = O_P((p^{1-\delta}s^{-1})^{1/2}h_n). \tag{7.11}$$

Next we consider $\|\widehat{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| \leq K_1 + K_2 + K_3 + K_4$, where

$$\begin{aligned} K_1 &= \|\widehat{\mathbf{A}}^T\mathbf{A}\widetilde{\mathbf{\Sigma}}_{\mathbf{x}}\mathbf{A}^T\widehat{\mathbf{A}} - \mathbf{\Sigma}_{\mathbf{x}}\|, \quad K_2 = \|\widehat{\mathbf{A}}^T\mathbf{A}\widetilde{\mathbf{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}}\widehat{\mathbf{A}}\|, \\ K_3 &= \|\widehat{\mathbf{A}}^T\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon},\mathbf{x}}\mathbf{A}^T\widehat{\mathbf{A}}\|, \qquad\qquad K_4 = \|\widehat{\mathbf{A}}^T(\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} - \widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}})\widehat{\mathbf{A}}\|, \end{aligned}$$

where $\widetilde{\mathbf{\Sigma}}_{\mathbf{x}} = n^{-1}\sum_{t=1}^{n}(\mathbf{x}_t - \bar{\mathbf{x}})(\mathbf{x}_t - \bar{\mathbf{x}})^T$. Now

$$\begin{aligned} K_1 &\leq \|\widehat{\mathbf{A}}^T\mathbf{A} - \mathbf{I}_r\| \cdot \|\widetilde{\mathbf{\Sigma}}_{\mathbf{x}}\| + \|\widetilde{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| \\ &= O_P(\|\widehat{\mathbf{A}}^T\mathbf{A} - \mathbf{I}_r\| \cdot (\|\widetilde{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| + \|\mathbf{\Sigma}_{\mathbf{x}}\|) + \|\widetilde{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\|) \\ &= O_P(p^{1-\delta}h_n), \end{aligned}$$

where we used $\|\widehat{\mathbf{A}}^T\mathbf{A} - \mathbf{I}_r\| = \|(\widehat{\mathbf{A}} - \mathbf{A})^T\mathbf{A}\| = O_P(h_n)$ by Theorem 1, $\|\mathbf{\Sigma}_{\mathbf{x}}\| = O(p^{1-\delta})$ from assumption in model (2.2), and $\|\widetilde{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| = O_P(p^{1-\delta}n^{-l_x})$ from Lemma 1. Next, using Lemma 1 and the fact that $\mathbf{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}} = \mathbf{0}$, we have

$$K_2 = O_P(K_3) = O_P(\|\widetilde{\mathbf{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}} - \mathbf{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}\|) = O_P(p^{1-\delta/2}n^{-l_{x\epsilon}}).$$

Finally, using Lemma 1 again and (7.11),

$$K_4 = O_P(\|\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}\| + \|\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}\|) = O_P((p^{1-\delta}s^{-1})^{1/2}h_n + pn^{-l_\epsilon}).$$

Looking at the rates for K_1 to K_4, and noting assumption (M2) and the definition of h_n, we can easily see that

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| = O_P(p^{1-\delta}h_n). \tag{7.12}$$

From (7.7), combining (7.11) and (7.12) and noting assumption (M2), the rate for $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}$ in the spectral norm is established, and the proof of the theorem completes. □

Proof of Theorem 3. We first show the rate for $\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}$. We use the standard inequality

$$\|M_1^{-1} - M_2^{-1}\| \leq \frac{\|M_2^{-1}\|^2 \cdot \|M_1 - M_2\|}{1 - \|(M_1 - M_2) \cdot M_2^{-1}\|}, \tag{7.13}$$

with $M_1 = \widetilde{\mathbf{\Sigma}}_{\mathbf{y}}$ and $M_2 = \mathbf{\Sigma}_{\mathbf{y}}$. Under assumption (M1) we have $\|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}\| \asymp 1 \asymp \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|$, so that

$$\begin{aligned}
\|\mathbf{\Sigma}_{\mathbf{y}}^{-1}\| &= \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}(\mathbf{\Sigma}_{\mathbf{x}}^{-1} + \mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\| \\
&= O(\|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\| + \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|^2 \cdot \|(\mathbf{\Sigma}_{\mathbf{x}}^{-1} + \mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\|) \\
&= O(1),
\end{aligned}$$

where we also used

$$\begin{aligned}
\|(\mathbf{\Sigma}_{\mathbf{x}}^{-1} + \mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\| &\leq \|(\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\| \\
&= \lambda_{\max}\{(\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\} = \lambda_{\min}^{-1}(\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}) = O(1),
\end{aligned} \tag{7.14}$$

since the eigenvalues of $\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}$ are of constant order by assumption (M1), and $\|\mathbf{A}\|_{\min} = 1$ with $\mathbf{A}\mathbf{x} \neq \mathbf{0}$ for any $\mathbf{x}$ since $\mathbf{A}$ is of full rank with $p > r$, so that

$$\begin{aligned}
\lambda_{\min}(\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}) = \min_{\mathbf{x}\neq\mathbf{0}} \frac{\mathbf{x}^T\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}\mathbf{x}}{\|\mathbf{x}\|} &\geq \min_{\mathbf{y}\neq\mathbf{0}} \frac{\mathbf{y}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{y}}{\|\mathbf{y}\|} \cdot \min_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} \\
&= \lambda_{\min}(\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}) \cdot \|\mathbf{A}\|_{\min} \asymp 1.
\end{aligned} \tag{7.15}$$

Then by (7.13) together with Theorem 2 that $\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| = O_P(p^{1-\delta}h_n) = o_P(1)$, we have

$$\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = \frac{O(1)^2 \cdot O_P(p^{1-\delta}h_n)}{1 - o_P(1)} = O_P(p^{1-\delta}h_n),$$

which is what we need to show.

Now we show the rate for $\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1}$. Using the Sherman-Morrison-Woodbury formula, we have $\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| \leq \sum_{j=1}^6 K_j$, where

$$\begin{aligned}
K_1 &= \|\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|, \\
K_2 &= \|(\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1})\widehat{\mathbf{A}}(\widehat{\mathbf{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1}\widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\|, \\
K_3 &= \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}}(\widehat{\mathbf{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1}\widehat{\mathbf{A}}^T(\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1})\|, \\
K_4 &= \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}(\widehat{\mathbf{A}} - \mathbf{A}_x)(\widehat{\mathbf{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1}\widehat{\mathbf{A}}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|, \\
K_5 &= \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}_x(\widehat{\mathbf{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1}(\widehat{\mathbf{A}} - \mathbf{A}_x)^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|, \\
K_6 &= \|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}_x\{(\widehat{\mathbf{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1} - (\mathbf{\Sigma}_{\mathbf{x}}^{-1} + \mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\}\mathbf{A}^T\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|.
\end{aligned} \tag{7.16}$$

First, we have $\|\mathbf{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\| = O(1)$ as before by assumption (M1). Next,

$$K_1 = \max_{1\leq j\leq k} |\widehat{\sigma}_j^{-2} - \sigma_j^{-2}| = \frac{\max_{1\leq j\leq k} |\widehat{\sigma}_j^2 - \sigma_j^2|}{\min_{1\leq j\leq k} \sigma_j^2(\sigma_j^2 + O_P((p^{1-\delta}s^{-1})^{1/2}h_n))} = O_P((p^{1-\delta}s^{-1})^{1/2}h_n),$$

where we used (7.11) and assumptions (M1) and (M2). From these, we have

$$\|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\| = O_P(1). \tag{7.17}$$

Also, like (7.14),

$$\|(\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}} + \widehat{\mathbf{A}}^T\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1}\| = O_P(1), \tag{7.18}$$

noting (7.11) and assumption (M1). With these rates and noting that $\|\mathbf{A}_x\| = \|\widehat{\mathbf{A}}\| = 1$ and $\|\widehat{\mathbf{A}} - \mathbf{A}_x\| = O_P(h_n)$ from Theorem 1, we have from (7.16) that

$$\begin{aligned} \|\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}^{-1} - \boldsymbol{\Sigma}_{\mathbf{y}}^{-1}\| &= O_P((p^{1-\delta}s^{-1})^{1/2}h_n + h_n) \\ &\quad + O_P(\|(\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}}^{-1} + \widehat{\mathbf{A}}^T\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}})^{-1} - (\boldsymbol{\Sigma}_{\mathbf{x}}^{-1} + \mathbf{A}^T\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A})^{-1}\|), \end{aligned} \tag{7.19}$$

where the last term is contributed from K_6. Using (7.14) and (7.18), and the inequality $\|M_1^{-1} - M_2^{-1}\| = O_P(\|M_1^{-1}\| \cdot \|M_1 - M_2\| \cdot \|M_2^{-1}\|)$, the rate for this term can be shown to be $O_P(L_1 + L_2)$, where

$$L_1 = \|\widehat{\boldsymbol{\Sigma}}_{\mathbf{x}}^{-1} - \boldsymbol{\Sigma}_{\mathbf{x}}^{-1}\|, \quad L_2 = \|\widehat{\mathbf{A}}^T\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\widehat{\mathbf{A}} - \mathbf{A}^T\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\mathbf{A}\|.$$

Consider $\|\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}\| \leq \|\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}\| = \lambda_{\min}^{-1}(\boldsymbol{\Sigma}_{\mathbf{x}}) = O(p^{-(1-\delta)})$ by assumption in model (2.2). With this and (7.12), substituting $M_1 = \widehat{\boldsymbol{\Sigma}}_{\mathbf{x}}$ and $M_2 = \boldsymbol{\Sigma}_{\mathbf{x}}$ into (7.13), we have

$$L_1 = \frac{O(p^{-(2-2\delta)}) \cdot O_P(p^{1-\delta}h_n)}{1 - O_P(p^{1-\delta}h_n \cdot p^{-(1-\delta)})} = \frac{O_P(p^{-(1-\delta)}h_n)}{1 - o_P(1)} = O_P(p^{-(1-\delta)}h_n). \tag{7.20}$$

For L_2, using $\|\widehat{\mathbf{A}}\| = \|\mathbf{A}\| = 1$, $\|\widehat{\mathbf{A}} - \mathbf{A}\| = O_P(h_n)$ from Theorem 1, the rate for K_1 shown before and (7.17), we have

$$L_2 = O_P(\|\widehat{\mathbf{A}} - \mathbf{A}\| \cdot \|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1}\| + \|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}^{-1} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}^{-1}\|) = O_P(h_n + (p^{1-\delta}s^{-1})^{1/2}h_n). \tag{7.21}$$

Hence, from (7.19), together with (7.20) and (7.21), we have

$$\|\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}}^{-1} - \boldsymbol{\Sigma}_{\mathbf{y}}^{-1}\| = O_P((1 + (p^{1-\delta}s^{-1})^{1/2})h_n),$$

which completes the proof of the theorem. □

Proof of Theorem 4. The idea of the proof is similar to that for Theorem 1 for the simple procedure. We want to find the order of the eigenvalues of the matrix $\mathbf{L}$ first.

From model (4.9), we have for $i = 1, 2$,

$$
\begin{aligned}
\|\mathbf{\Sigma}_{ii}(k)\|^2 &\asymp p^{2-2\delta_i} \asymp \|\mathbf{\Sigma}_{ii}(k)\|_{\min}, \\
\|\mathbf{\Sigma}_{12}(k)\|^2 &\asymp p^{2-\delta_1-\delta_2} \asymp \|\mathbf{\Sigma}_{12}(k)\|_{\min}, \\
\|\mathbf{\Sigma}_{i\boldsymbol{\epsilon}}(k)\|^2 &\asymp p^{2-\delta_i} \asymp \|\mathbf{\Sigma}_{i\boldsymbol{\epsilon}}(k)\|.
\end{aligned} \tag{7.22}
$$

We want to find the lower bounds of the order of the r_1-th largest eigenvalue, as well as the smallest non-zero eigenvalue of $\mathbf{L}$. We first note that

$$
\begin{aligned}
\mathbf{\Sigma}_{\mathbf{y}}(k) = \mathbf{A}_1\mathbf{\Sigma}_{11}(k)\mathbf{A}_1^T + \mathbf{A}_1\mathbf{\Sigma}_{12}(k)\mathbf{A}_2^T + \mathbf{A}_2\mathbf{\Sigma}_{21}(k)\mathbf{A}_1^T \\
+ \mathbf{A}_2\mathbf{\Sigma}_{22}(k)\mathbf{A}_2^T + \mathbf{A}_1\mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k) + \mathbf{A}_2\mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k),
\end{aligned} \tag{7.23}
$$

and hence

$$
\mathbf{L} = \mathbf{A}_1\mathbf{W}_1\mathbf{A}_1^T + \mathbf{A}_2\mathbf{W}_2\mathbf{A}_2^T + \text{cross terms}, \tag{7.24}
$$

where $\mathbf{W}_1$ (with size $r_1 \times r_1$) and $\mathbf{W}_2$ (with size $r_2 \times r_2$) are positive semi-definite matrices defined by

$$
\begin{aligned}
\mathbf{W}_1 &= \sum_{k=1}^{k_0} \Big\{ \mathbf{\Sigma}_{11}(k)\mathbf{\Sigma}_{11}(k)^T + \mathbf{\Sigma}_{12}(k)\mathbf{\Sigma}_{12}(k)^T + \mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k)\mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k)^T \Big\}, \\
\mathbf{W}_2 &= \sum_{k=1}^{k_0} \Big\{ \mathbf{\Sigma}_{21}(k)\mathbf{\Sigma}_{21}(k)^T + \mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)\mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)^T + \mathbf{\Sigma}_{22}(k)\mathbf{\Sigma}_{22}(k)^T \Big\}.
\end{aligned}
$$

From $\mathbf{W}_1$, by (7.22) and that $\|\mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k)\| = o(\|\mathbf{\Sigma}_{11}(k)\|)$, we have the order of the r_1 eigenvalues for $\mathbf{W}_1$ is all $p^{2-2\delta_1}$. Then the r_1-th largest eigenvalue of $\mathbf{L}$ is of order $p^{2-2\delta_1}$ since the term $\mathbf{\Sigma}_{11}(k)\mathbf{\Sigma}_{11}(k)^T$ has the largest order at $p^{2-2\delta_1}$. We write

$$
p^{2-2\delta_1} = O_P(\lambda_{r_1}(\mathbf{L})), \tag{7.25}
$$

where $\lambda_i(M)$ represents the i-th largest eigenvalue of the square matrix M.

For the smallest non-zero eigenvalue of $\mathbf{W}_2$, since $\|\mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| = o(\|\mathbf{\Sigma}_{22}(k)\|)$, it is contributed either from the term $\mathbf{\Sigma}_{22}(k)\mathbf{\Sigma}_{22}(k)^T$ or $\mathbf{\Sigma}_{21}(k)\mathbf{\Sigma}_{21}(k)^T$ in $\mathbf{W}_2$, and has order $p^{2-2\delta_2}$ if $\|\mathbf{\Sigma}_{21}(k)\|_{\min} = o(p^{2-2\delta_2})$, and p^{2-c} in general if $\|\mathbf{\Sigma}_{21}(k)\|_{\min} \asymp p^{2-c}$, with $\delta_1+\delta_2 \le c \le 2\delta_2$. Hence

$$
\begin{aligned}
p^{2-2\delta_2} &= O_P(\lambda_{r_1+r_2}(\mathbf{L})), \quad \text{if } \|\mathbf{\Sigma}_{21}(k)\|_{\min} = o(p^{2-2\delta_2}), \\
p^{2-c} &= O_P(\lambda_{r_1+r_2}(\mathbf{L})), \quad \text{if } \|\mathbf{\Sigma}_{21}(k)\|_{\min} \asymp p^{2-c},\ \delta_1 + \delta_2 \le c < 2\delta_2.
\end{aligned} \tag{7.26}
$$

Now we can write $\mathbf{L} = (\mathbf{A}\ \mathbf{B})\mathbf{D}_x(\mathbf{A}\ \mathbf{B})^T$, where $\mathbf{B}$ is the orthogonal complement of $\mathbf{A}$, and with $\mathbf{D}_{x1}$ containing the r_1 largest eigenvalues of $\mathbf{L}$ and $\mathbf{D}_{x2}$ the next r_2 largest,

$$\mathbf{D}_x = \begin{pmatrix} \mathbf{D}_{x1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_{x2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix}, \tag{7.27}$$

with $p^{2-2\delta_1} = O_P(\lambda_{\min}(\mathbf{D}_{x1}))$ by (7.25), and $p^{2-c} = O_P(\lambda_{\min}(\mathbf{D}_{x2}))$ by (7.26), $\delta_1 + \delta_2 \leq c \leq 2\delta_2$.

Similar to the proof of Theorem 1, we define $\mathbf{E_L} = \widetilde{\mathbf{L}} - \mathbf{L}$. Then (7.3) holds, and

$$\|\mathbf{\Sigma_y}(k)\| = O_P(p^{1-\delta_1}), \tag{7.28}$$

using (7.23) and $\|\mathbf{\Sigma}_{11}(k)\| = O_P(p^{1-\delta_1})$ from (7.22).

Also,

$$\begin{aligned} \|\widetilde{\mathbf{\Sigma}}_\mathbf{y}(k) - \mathbf{\Sigma_y}(k)\| &= O_P(\|\widetilde{\mathbf{\Sigma}}_{11}(k) - \mathbf{\Sigma}_{11}(k)\| + \|\widetilde{\mathbf{\Sigma}}_{12}(k) - \mathbf{\Sigma}_{12}(k)\| \\ &\quad + \|\widetilde{\mathbf{\Sigma}}_{22}(k) - \mathbf{\Sigma}_{22}(k)\| + \|\widetilde{\mathbf{\Sigma}}_{1\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k)\| \\ &\quad + \|\widetilde{\mathbf{\Sigma}}_{2\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| + \|\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}(k)\|) \\ &= O_P(p^{1-\delta_1}n^{-l_1} + p^{1-\delta_2}n^{-l_2} + p^{1-\delta_1/2-\delta_2/2}n^{-l_{12}} \\ &\quad + p^{1-\delta_1/2}n^{-l_{1\epsilon}} + p^{1-\delta_2/2}n^{-l_{2\epsilon}} + pn^{-l_\epsilon}) = O_P(p^{1-\delta_1}\omega_1), \end{aligned} \tag{7.29}$$

where we used condition (D') in section 4.2 to derive the following rates like those in Lemma 1 (proofs thus omitted):

$$\begin{aligned} &\|\widetilde{\mathbf{\Sigma}}_{11}(k) - \mathbf{\Sigma}_{11}(k)\| = O_P(p^{1-\delta_1}n^{-l_1}), \quad \|\widetilde{\mathbf{\Sigma}}_{12}(k) - \mathbf{\Sigma}_{12}(k)\| = O_P(p^{1-\delta_1/2-\delta_2/2}n^{-l_{12}}), \\ &\|\widetilde{\mathbf{\Sigma}}_{22}(k) - \mathbf{\Sigma}_{22}(k)\| = O_P(p^{1-\delta_2}n^{-l_2}), \quad \|\widetilde{\mathbf{\Sigma}}_{1\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{1\boldsymbol{\epsilon}}(k)\| = O_P(p^{1-\delta_1/2}n^{-l_{1\epsilon}}), \\ &\|\widetilde{\mathbf{\Sigma}}_{2\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| = O_P(p^{1-\delta_2/2}n^{-l_{2\epsilon}}), \quad \|\widetilde{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}}(k) - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}(k)\| = O_P(pn^{-l_\epsilon}). \end{aligned} \tag{7.30}$$

We form $\widehat{\mathbf{A}}_1$ with the first r_1 unit eigenvectors corresponding to the r_1 largest eigenvalues, i.e. the eigenvalues in $\mathbf{D}_{x1}$. Now, we have

$$\begin{aligned} \|\mathbf{E_L}\| &= O_P(\|\widetilde{\mathbf{\Sigma}}_\mathbf{y}(k) - \mathbf{\Sigma_y}(k)\| \cdot (\|\mathbf{\Sigma_y}(k)\| + \|\widetilde{\mathbf{\Sigma}}_\mathbf{y}(k) - \mathbf{\Sigma_y}(k)\|)) = O_P(p^{2-2\delta_1}\omega_1) \\ &= o_P(p^{2-2\delta_1}) = O_P(\lambda_{\min}(\mathbf{D}_{x1})), \text{ hence asymptotically,} \\ \|\mathbf{E_L}\| &\leq \frac{1}{5}\mathrm{sep}\left(\mathbf{D}_{x1}, \begin{pmatrix} \mathbf{D}_{x2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\right), \end{aligned}$$

where the second equality is from (7.28) and (7.29), the third is from noting that $\omega_1 = o(1)$, and the last is from (7.25). Hence, we can use Lemma 2 and arguments similar to the proof of Theorem 1 to conclude that

$$\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P\Bigg(\|\mathbf{E_L}\|/\text{sep}\Bigg(\mathbf{D}_{x1}, \begin{pmatrix} \mathbf{D}_{x2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\Bigg)\Bigg) = O_P(\omega_1).$$

Similarly, depending on the order of $\|\mathbf{\Sigma}_{21}(k)\|_{\min}$, we have

$$\|\mathbf{E_L}\| = O_P(p^{2-2\delta_1}\omega_1) \leq \frac{1}{5}\text{sep}\Bigg(\mathbf{D}_{x2}, \begin{pmatrix} \mathbf{D}_{x1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\Bigg) \asymp p^{2-c},$$

since we assumed $p^{c-2\delta_1}\omega_1 = o(1)$ for $\delta_1 + \delta_2 \leq c \leq 2\delta_2$. Hence

$$\|\widehat{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P\Bigg(\|\mathbf{E_L}\|/\text{sep}\Bigg(\mathbf{D}_{x2}, \begin{pmatrix} \mathbf{D}_{x1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\Bigg)\Bigg) = O_P(p^{c-2\delta_1}\omega_1) = O_P(\omega_2).$$

This completes the proof for the simple procedure.

For the two-step procedure, denote $\mathbf{y}_t^* = (\mathbf{I}_p - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T)\mathbf{y}_t$, and define $\mathbf{E}_{\mathbf{L}^*} = \widetilde{\mathbf{L}}^* - \mathbf{L}^*$. Note that

$$\|\mathbf{E}_{\mathbf{L}^*}\| = \|\widetilde{\mathbf{L}}^* - \mathbf{L}^*\| \leq \sum_{k=1}^{k_0}\Big\{\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}^*}(k) - \mathbf{\Sigma}_{\mathbf{y}^*}(k)\|^2 + \|\mathbf{\Sigma}_{\mathbf{y}^*}(k)\|\cdot\|\widetilde{\mathbf{\Sigma}}_{\mathbf{y}^*}(k) - \mathbf{\Sigma}_{\mathbf{y}^*}(k)\|\Big\}, \quad (7.31)$$

where

$$\begin{aligned}
\widetilde{\mathbf{\Sigma}}_{\mathbf{y}^*}(k) &= (\mathbf{I}_p - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T)\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}(k)(\mathbf{I}_p - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T), \\
\mathbf{\Sigma}_{\mathbf{y}^*}(k) &= (\mathbf{I}_p - \mathbf{A}_1\mathbf{A}_1^T)\mathbf{\Sigma}_{\mathbf{y}}(k)(\mathbf{I}_p - \mathbf{A}_1\mathbf{A}_1^T) = \mathbf{A}_2\mathbf{\Sigma}_{22}(k)\mathbf{A}_2^T + \mathbf{A}_2\mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)(\mathbf{I}_p - \mathbf{A}_1\mathbf{A}_1^T), \\
\widetilde{\mathbf{L}}^* &= \sum_{k=1}^{k_0}\widetilde{\mathbf{\Sigma}}_{\mathbf{y}^*}(k)\widetilde{\mathbf{\Sigma}}_{\mathbf{y}^*}(k)^T, \quad \mathbf{L}^* = \sum_{k=1}^{k_0}\mathbf{\Sigma}_{\mathbf{y}^*}(k)\mathbf{\Sigma}_{\mathbf{y}^*}(k)^T,
\end{aligned}$$

with $\widehat{\mathbf{A}}_1$ being the estimator from the simple procedure, so that $\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(\omega_1)$ from previous result. We write

$$\mathbf{L}^* = \mathbf{A}_2\mathbf{Q}_2\mathbf{D}_2\mathbf{Q}_2^T\mathbf{A}_2^T, \text{ so that } \mathbf{L}^*\mathbf{A}_2\mathbf{Q}_2 = \mathbf{A}_2\mathbf{Q}_2\mathbf{D}_2,$$

and hence we are estimating $\mathbf{A}_2' = \mathbf{A}_2\mathbf{Q}_2$.

The idea of the proof is to find the rates of $\|\mathbf{E}_{\mathbf{L}^*}\|$ and the eigenvalues in $\mathbf{D}_2$ and use the arguments similar to the proof for the simple procedure to get the rate for $\|\widehat{\mathbf{A}}_2' - \mathbf{A}_2'\|$.

First, with the assumption that $\|\mathbf{\Sigma}_{22}(k)\| \asymp p^{1-\delta_2} \asymp \|\mathbf{\Sigma}_{22}(k)\|_{\min}$ and $\|\mathbf{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| = o(\|\mathbf{\Sigma}_{22}(k)\|)$, all the eigenvalues in $\mathbf{D}_2$ have order $p^{2-2\delta_2}$.

We need to find $\|\mathbf{E}_{\mathbf{L}^*}\|$. It is easy to show that

$$
\begin{aligned}
\|\widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T\mathbf{A}_2\| = \|\widehat{\mathbf{A}}_1(\widehat{\mathbf{A}}_1 - \mathbf{A}_1)^T\mathbf{A}_2\| \le \|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(\omega_1), \\
\|(\mathbf{I}_p - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T)\mathbf{A}_1\| = \|(\mathbf{A}_1 - \widehat{\mathbf{A}}_1) - \widehat{\mathbf{A}}_1(\widehat{\mathbf{A}}_1 - \mathbf{A}_1)^T\mathbf{A}_1\| = O_P(\omega_1).
\end{aligned} \tag{7.32}
$$

Writing $\widehat{\mathbf{H}}_1 = \mathbf{I}_p - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T$, we can decompose $\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}^*}(k) - \boldsymbol{\Sigma}_{\mathbf{y}^*}(k) = \sum_{l=1}^{9} I_l$, where

$$
\begin{aligned}
& I_1 = \widehat{\mathbf{H}}_1\mathbf{A}_1\widetilde{\boldsymbol{\Sigma}}_{11}(k)\mathbf{A}_1^T\widehat{\mathbf{H}}_1, \quad I_2 = \widehat{\mathbf{H}}_1\mathbf{A}_1\widetilde{\boldsymbol{\Sigma}}_{12}(k)\mathbf{A}_2^T\widehat{\mathbf{H}}_1, \\
& I_3 = \widehat{\mathbf{H}}_1\mathbf{A}_1\widetilde{\boldsymbol{\Sigma}}_{1\boldsymbol{\epsilon}}(k)\widehat{\mathbf{H}}_1, \qquad I_4 = \widehat{\mathbf{H}}_1\mathbf{A}_2\widetilde{\boldsymbol{\Sigma}}_{21}(k)\mathbf{A}_1^T\widehat{\mathbf{H}}_1, \\
& I_5 = \widehat{\mathbf{H}}_1\mathbf{A}_2\widetilde{\boldsymbol{\Sigma}}_{22}(k)\mathbf{A}_2^T\widehat{\mathbf{H}}_1 - \mathbf{A}_2\boldsymbol{\Sigma}_{22}(k)\mathbf{A}_2^T, \quad I_6 = \widehat{\mathbf{H}}_1\mathbf{A}_2\widetilde{\boldsymbol{\Sigma}}_{2\boldsymbol{\epsilon}}(k)\widehat{\mathbf{H}}_1 - \mathbf{A}_2\boldsymbol{\Sigma}_{2\boldsymbol{\epsilon}}(k)\mathbf{H}_1, \\
& I_7 = \widehat{\mathbf{H}}_1\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}1}(k)\mathbf{A}_1^T\widehat{\mathbf{H}}_1, \quad I_8 = \widehat{\mathbf{H}}_1\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}2}(k)\mathbf{A}_2^T\widehat{\mathbf{H}}_1, \quad I_9 = \widehat{\mathbf{H}}_1\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}(k)\widehat{\mathbf{H}}_1.
\end{aligned}
$$

Using (7.30), (7.32) and the assumptions in model (4.9), we can see that

$$
\begin{aligned}
& \|I_1\| = O_P(\|\widehat{\mathbf{H}}_1\mathbf{A}_1\|^2 \cdot \|\widetilde{\boldsymbol{\Sigma}}_{11}(k)\|) = O_P(p^{1-\delta_1}\omega_1^2), \\
& \|I_2\| = O_P(\|\widehat{\mathbf{H}}_1\mathbf{A}_1\| \cdot \|\widetilde{\boldsymbol{\Sigma}}_{12}(k)\|) = O_P(p^{1-\delta_1/2-\delta_2/2}\omega_1) = \|I_4\|, \\
& \|I_7\| = O_P(\|I_3\|) = O_P(\|\widehat{\mathbf{H}}_1\mathbf{A}_1\| \cdot (\|\widetilde{\boldsymbol{\Sigma}}_{1\boldsymbol{\epsilon}}(k) - \boldsymbol{\Sigma}_{1\boldsymbol{\epsilon}}(k)\| + \|\boldsymbol{\Sigma}_{1\boldsymbol{\epsilon}}(k)\|)) = O_P(p^{1-\delta_1}\omega_1), \\
& \|I_6\| = O_P(\|\widetilde{\boldsymbol{\Sigma}}_{2\boldsymbol{\epsilon}}(k) - \boldsymbol{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| + \|\boldsymbol{\Sigma}_{2\boldsymbol{\epsilon}}(k)\| \cdot \|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\|) = O_P(p^{1-\delta_2/2}(n^{-l_{2\epsilon}} + p^{-\delta_2/2}\omega_1)), \\
& \|I_8\| = O_P(\|\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}2}(k)\|) = O_P(p^{1-\delta_2/2}n^{-l_{2\epsilon}}), \quad \|I_9\| = O_P(\|\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}}\|) = O_P(pn^{-l_\epsilon}), \\
& \|I_5\| = O_P(\|\widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1^T\mathbf{A}_2\| \cdot \|\widetilde{\boldsymbol{\Sigma}}_{22}(k)\| + \|\widetilde{\boldsymbol{\Sigma}}_{22}(k) - \boldsymbol{\Sigma}_{22}(k)\|) = O_P(p^{1-\delta_2}(\omega_1 + n^{-l_2})).
\end{aligned}
$$

Hence, we have

$$
\begin{aligned}
\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}^*}(k) - \boldsymbol{\Sigma}_{\mathbf{y}^*}(k)\| &= O_P(p^{1-\delta_1}\omega_1^2 + p^{1-\delta_1/2-\delta_2/2}\omega_1 + p^{1-\delta_1}\omega_1 \\
&\quad + p^{1-\delta_2/2}(n^{-l_{2\epsilon}} + p^{-\delta_2/2}\omega_1) + pn^{-l_\epsilon} + p^{1-\delta_2}(\omega_1 + n^{-l_2})) \\
&= O_P(p^{1-\delta_1}\omega_1).
\end{aligned} \tag{7.33}
$$

We also have

$$
\|\boldsymbol{\Sigma}_{\mathbf{y}^*}(k)\| = O_P(\|\boldsymbol{\Sigma}_{22}(k)\| + \|\boldsymbol{\Sigma}_{2\boldsymbol{\epsilon}}(k)\|) = O_P(p^{1-\delta_2}). \tag{7.34}
$$

Hence, with (7.33) and (7.34), (7.31) becomes

$$
\|\mathbf{E}_{\mathbf{L}^*}\| = O_P(p^{2-\delta_1-\delta_2}\omega_1). \tag{7.35}
$$

With the order of eigenvalues in $\mathbf{D}_2$ being $p^{2-2\delta_2}$ and (7.33), we can use Lemma 2 and the arguments similar to those in the proof of Theorem 1 to get

$$
\|\widehat{\mathbf{A}}_2' - \mathbf{A}_2'\| = O_P(\|\mathbf{E}_{\mathbf{L}^*}\|/\mathrm{sep}(\mathbf{D}_2, \mathbf{0})) = O_P(p^{\delta_2-\delta_1}\omega_1),
$$

and the proof of the theorem completes. $\square$

Proof of Theorem 5. We have $\widehat{\mathbf{x}}_t = \widehat{\mathbf{A}}^T\mathbf{y}_t = \widehat{\mathbf{A}}^T\mathbf{A}\mathbf{x}_t + (\widehat{\mathbf{A}} - \mathbf{A})^T\boldsymbol{\epsilon}_t + \mathbf{A}^T\boldsymbol{\epsilon}_t$. With $\mathbf{A} = (\mathbf{A}_1\ \mathbf{A}_2)$ and $\mathbf{x}_t = (\mathbf{x}_{1t}^T\ \mathbf{x}_{2t}^T)^T$, we have

$$\begin{aligned}
\widehat{\mathbf{x}}_{1t} &= \widehat{\mathbf{A}}_1^T\mathbf{A}_1\mathbf{x}_{1t} + \widehat{\mathbf{A}}_1^T\mathbf{A}_2\mathbf{x}_{2t} + (\widehat{\mathbf{A}}_1 - \mathbf{A}_1)^T\boldsymbol{\epsilon}_t + \mathbf{A}_1^T\boldsymbol{\epsilon}_t,\\
\widehat{\mathbf{x}}_{2t} &= \widehat{\mathbf{A}}_2^T\mathbf{A}_1\mathbf{x}_{1t} + \widehat{\mathbf{A}}_2^T\mathbf{A}_2\mathbf{x}_{2t} + (\widehat{\mathbf{A}}_2 - \mathbf{A}_2)^T\boldsymbol{\epsilon}_t + \mathbf{A}_2^T\boldsymbol{\epsilon}_t.
\end{aligned}$$

We first note that for $i = 1, 2$, $\|(\widehat{\mathbf{A}}_i - \mathbf{A}_i)^T\boldsymbol{\epsilon}_t\| = O_P(\mathbf{A}_i^T\boldsymbol{\epsilon}_t) = O_P(1)$ since $\|\widehat{\mathbf{A}}_i - \mathbf{A}_i\| = o_P(1)$ and $\mathbf{A}_i^T\boldsymbol{\epsilon}_t$ are r_i $O_P(1)$ random variables. Then

$$\begin{aligned}
\|\widehat{\mathbf{x}}_{1t}\| = \|\widehat{\mathbf{x}}_{1t}\|_F &\geq \|\widehat{\mathbf{A}}_1^T\mathbf{A}_1\mathbf{x}_{1t}\|_F - \|\widehat{\mathbf{A}}_1^T\mathbf{A}_2\mathbf{x}_{2t}\|_F + O_P(1)\\
&\geq \|\widehat{\mathbf{A}}_1^T\mathbf{A}_1\|_{\min}\cdot\|\mathbf{x}_{1t}\|_F - \|\widehat{\mathbf{A}}_1^T\mathbf{A}_2\|\cdot\|\mathbf{x}_{2t}\| + O_P(1)\\
&\geq \|\widehat{\mathbf{A}}_1\|_{\min}\cdot\|\mathbf{A}_1\|_{\min}\cdot\|\mathbf{x}_{1t}\| - o_P(\|\mathbf{x}_{2t}\|) + O_P(1)\\
&= O_P(\|\mathbf{x}_{1t}\|) \asymp p^{\frac{1-\delta_1}{2}},
\end{aligned}$$

where $\|M\|_F$ denotes the Frobenius norm of the matrix M, and we used the inequality $\|\mathbf{AB}\|_F \geq \|\mathbf{A}\|_{\min}\cdot\|\mathbf{B}\|_F$. Finally, with similar arguments,

$$\begin{aligned}
\|\widehat{\mathbf{x}}_{2t}\| &\leq \|\mathbf{x}_{2t}\| + \|\widehat{\mathbf{A}}_2^T\mathbf{A}_2\|\cdot\|\mathbf{x}_{1t}\| + O_P(1)\\
&= O_P(p^{\frac{1-\delta_2}{2}}) + O_P(\|\widehat{\mathbf{A}}_2 - \mathbf{A}_2\|\cdot p^{\frac{1-\delta_1}{2}})\\
&= O_P(p^{\frac{1-\delta_2}{2}}) + o_P(p^{\frac{1-\delta_1}{2}})\\
&= o_P(p^{\frac{1-\delta_1}{2}}),
\end{aligned}$$

which establishes the claim of the theorem. $\square$

Proof of Theorem 6. We can easily use the decomposition in (7.6) again for model (4.9) to arrive at

$$\begin{aligned}
\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{y}} - \boldsymbol{\Sigma}_{\mathbf{y}}\| &= O_P(p^{1-\delta_1}n^{-l_1} + p^{1-\delta_2}n^{-l_2} + p^{1-\delta_1/2}n^{-l_{1\epsilon}} + p^{1-\delta_2/2}n^{-l_{2\epsilon}} + pn^{-l_\epsilon}),\\
&= O_P(p^{1-\delta_1}\omega_1),
\end{aligned}$$

where we used assumption (D'), and arguments like those in Lemma 1 to arrive at $\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x}} - \boldsymbol{\Sigma}_{\mathbf{x}}\| = O_P(p^{1-\delta_1}n^{-l_1} + p^{1-\delta_2}n^{-l_2})$, $\|\widetilde{\boldsymbol{\Sigma}}_{\mathbf{x},\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\mathbf{x},\boldsymbol{\epsilon}}\| = O_P(p^{1-\delta_1/2}n^{-l_{1\epsilon}} + p^{1-\delta_2/2}n^{-l_{2\epsilon}})$ and $\|\widetilde{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\| = O_P(pn^{-l_\epsilon})$.

Now consider $\|\widehat{\boldsymbol{\Sigma}}_{\mathbf{y}} - \boldsymbol{\Sigma}_{\mathbf{y}}\|$, which can be decomposed like that in (7.7). Hence we need to consider $\|\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\epsilon}} - \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}\| = \max_{1\leq j\leq p}|\widehat{\sigma}_j^2 - \sigma_j^2| \leq I_1 + I_2 + I_3$, where

$$I_1 = O_P(p^{1-\delta_1}s^{-1}\|\widehat{\mathbf{A}} - \mathbf{A}\|^2),$$

which used decomposition in (7.8), and $\|\mathbf{X}\|_F^2 = O_P(p^{1-\delta_1}nr_1 + p^{1-\delta_2}nr_2) = O_P(p^{1-\delta_1}n)$;

$$I_2 = O_P(n^{-l_\epsilon} + s^{-1}),$$

where derivation is similar to that in (7.9) and thereafter. Also, $I_3 = O_P(I_1^{1/2}) = O_P((p^{1-\delta_1}s^{-1})^{1/2}\|\widehat{\mathbf{A}} - \mathbf{A}\|)$. Thus, with assumption (M2)', we see that

$$\|\widehat{\mathbf{\Sigma}}_{\boldsymbol{\epsilon}} - \mathbf{\Sigma}_{\boldsymbol{\epsilon}}\| = O_P((p^{1-\delta_1}s^{-1})^{1/2}\|\widehat{\mathbf{A}} - \mathbf{A}\|). \tag{7.36}$$

For $\|\widehat{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\|$, we use the decomposition like that in the proof of Theorem 2, and noting that $\|\mathbf{\Sigma}_{\mathbf{x}}\| = O(p^{1-\delta_1} + p^{1-\delta_2}) = O(p^{1-\delta_1})$, to arrive at

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{x}} - \mathbf{\Sigma}_{\mathbf{x}}\| = O_P(p^{1-\delta_1}\|\widehat{\mathbf{A}} - \mathbf{A}\|). \tag{7.37}$$

Hence noting assumption (M2)' again and combining (7.36) and (7.37), we see that

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}} - \mathbf{\Sigma}_{\mathbf{y}}\| = O_P(p^{1-\delta_1}\|\widehat{\mathbf{A}} - \mathbf{A}\|),$$

which completes the proof of the theorem. □

Proof of Theorem 7. We omit the rate for $\widetilde{\mathbf{\Sigma}}_{\mathbf{y}}$ since it involves standard treatments like that in Theorem 3.

Note that (7.19) becomes

$$\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = O_P((1 + (p^{1-\delta_1}s^{-1})^{1/2})\|\widehat{\mathbf{A}} - \mathbf{A}\|) + O_P(L_1 + L_2),$$

with L_1 and L_2 defined similar to those in the proof of Theorem 3. Similar to (7.20) and (7.21), we have respectively

$$L_1 = O_P(p^{-(1-\delta_2)}\|\widehat{\mathbf{A}} - \mathbf{A}\|), \quad L_2 = O_P((1 + (p^{1-\delta_1}s^{-1})^{1/2})\|\widehat{\mathbf{A}} - \mathbf{A}\|),$$

which shows that $\|\widehat{\mathbf{\Sigma}}_{\mathbf{y}}^{-1} - \mathbf{\Sigma}_{\mathbf{y}}^{-1}\| = O_P((1 + (p^{1-\delta_1}s^{-1})^{1/2})\|\widehat{\mathbf{A}} - \mathbf{A}\|)$. This completes the proof of the theorem. □

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